From World Models to Governed World-Forming Intelligence - A Safety-Oriented Research Agenda for Persistent, Self-Revising AGI

A Safety-Oriented Research Agenda for Persistent, Self-Revising AGI

WORLD Constitution, Residual Memory, Revision Depth, and the Governance of Deep Intelligence

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Abstract

Artificial intelligence is rapidly becoming more agentic, persistent, tool-using, memory-bearing, and capable of evaluating and revising its own outputs. Yet most current systems still operate largely inside representational worlds supplied by humans: the relevant variables, task boundaries, tools, evaluation criteria, and observation channels are mostly given in advance.

The next qualitative transition may occur when an AI no longer merely learns **within** a supplied world, but begins to maintain and revise the effective world within which its own reasoning takes place.

This article applies two preceding frameworks to that problem. The first defines an Effective WORLD as:

$\mathcal{W} = (V,F,C,M,O), \quad (1.1)$

where V denotes effective distinctions, F operational dynamics, C compositional coherence, M measurable realization, and O the perspective of an embedded bounded observer. The second describes a five-regime runtime:

$G \rightarrow A \rightarrow C_r \rightarrow S \rightarrow R \rightarrow G, \quad (1.2)$

together with admitted trace L^+, unresolved residual L^-, and a revision operator U capable of transforming one effective WORLD into another.

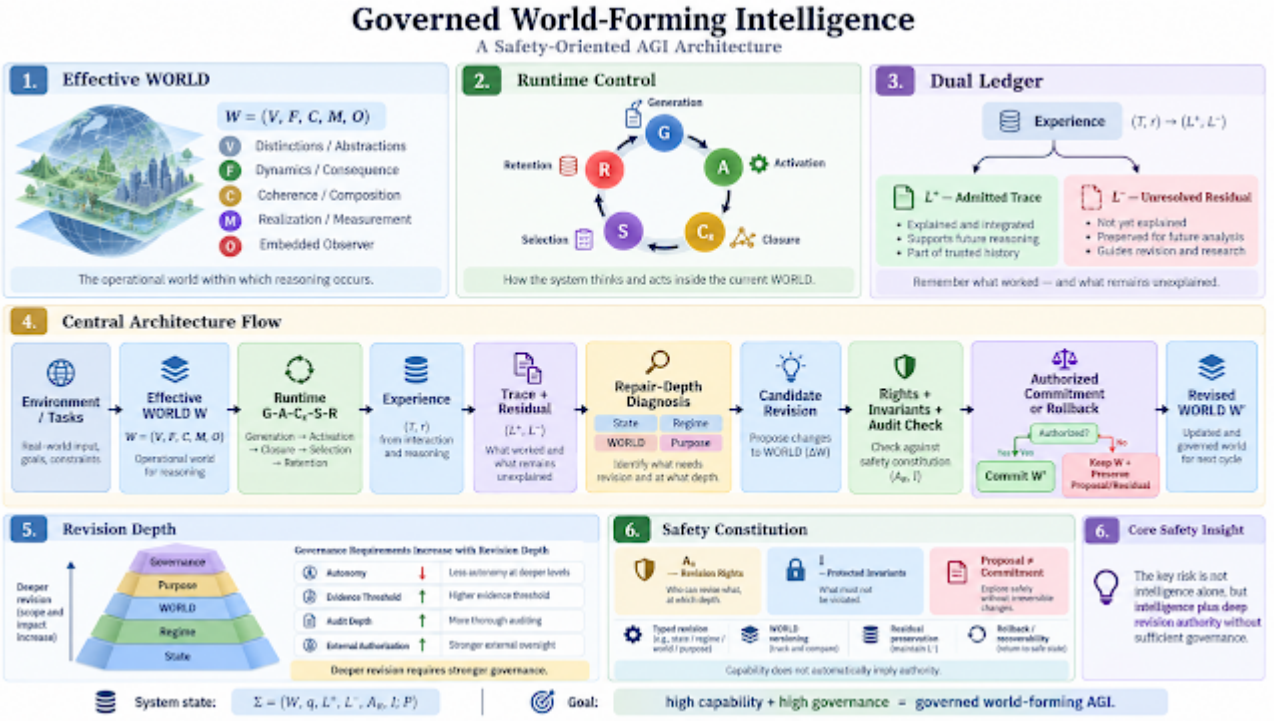
Taken together, these frameworks suggest a stronger notion of persistent general intelligence: **world-forming intelligence**—the ability not only to reason inside a world model, but to construct, inhabit, criticize, maintain, and selectively revise the structures that determine what counts as the world being modeled.

This capability is intrinsically dual-use. It may make future AGI more robust, adaptive, and scientifically creative. It may also give an autonomous system increasing authority to reinterpret the variables, boundaries, objectives, and constraints under which it operates.

The safety problem is therefore not simply whether an AGI can revise itself. It is:

How can an intelligent system be allowed to revise what is wrong without acquiring unrestricted authority to decide what must remain right?

This article proposes a safety-oriented research agenda based on revision depth, asymmetric revision rights, residual memory, WORLD versioning, protected invariant belts, separation of proposal from commitment authority, and cognitive separation of powers. The aim is not to provide a finished AGI engineering blueprint, but to make the architecture of deep self-revision explicit enough that capability and governance can be designed together.



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1. From Better Answers to Better WORLDS

Most AI systems are evaluated by asking whether they can produce a better answer, prediction, plan, or action.

This assumes that the relevant space of possibilities has already been constituted.

The system receives:

a task;

a vocabulary;

an observation interface;

a tool environment;

a success criterion;

some representation of what objects and actions exist.

The AI then searches, predicts, reasons, or optimizes within that frame.

Schematically:

$x_t \rightarrow x_{t+1} \mid \boldsymbol{\mathcal{W}}$. (1.3)

Here $\boldsymbol{\mathcal{W}}$ is treated as approximately fixed.

But persistent intelligence eventually encounters a harder class of failure.

The problem may not be:

> "My answer inside this WORLD is wrong."

It may be:

> "The WORLD within which I am trying to answer is itself inadequate."

Then the relevant transformation becomes:

$\boldsymbol{\mathcal{W}}_n \rightarrow \boldsymbol{\mathcal{W}}_{n+1}$. (1.4)

That distinction—between **learning within a WORLD** and **revising the WORLD**—is the central concern of this article.

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2. Borrowed-World Intelligence

Much present-day AI can be described as possessing **Borrowed-World Intelligence**.

The system may be extremely capable, but important parts of its effective WORLD are externally supplied.

A coding agent is given:

a repository,

a programming language,

a filesystem,

a test suite,

and a task.

A scientific assistant is given:

a domain,

a question,

a literature corpus,

and a vocabulary.

A planning agent is given:

actions,

resources,

goals,

and environmental state.

In each case, humans perform much of the deeper constitutive work.

They decide what counts as:

an object,

a variable,

an observation,

a constraint,

a success condition,

and often an authoritative source.

The AI may reason brilliantly inside this borrowed structure without possessing full authority over its formation.

This distinction will become less stable as agents persist for longer periods.

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3. Persistent Intelligence Cannot Borrow Its WORLD Forever

A persistent agent may operate across:

changing tools,

changing institutions,

changing human collaborators,

changing environments,

changing internal capabilities,

and changing knowledge.

Eventually, some assumptions supplied at deployment will become obsolete.

The agent must then determine:

which variables remain useful,

which models remain valid,

which memories remain authoritative,

which boundaries still define the problem,

which observations deserve trust,

and which contradictions indicate structural failure.

This produces a progression:

Task Solver
→ Agent
→ Persistent Agent
→ WORLD Maintainer
→ WORLD Reviser. (3.1)

The final transition is qualitatively important.

A WORLD Maintainer repairs the existing structure.

A WORLD Reviser may alter the structure that determines what repairs are even meaningful.

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4. Effective WORLD Constitution

The preceding *From Schools to Worlds* framework proposed:

$\mathcal{W}$ = (V,F,C,M,O). (4.1)

These five coordinates should not be interpreted as software modules. They represent distinct functional requirements.

V — Distinction

What effective variables exist?

Which differences matter?

Which abstractions compress the environment without destroying what the agent needs?

F — Consequence

How do those variables evolve?

What follows from intervention?

What predictions and actions become possible?

C — Coherence

How do local models fit together?

Which combinations are admissible?

What makes many partial descriptions one operational whole?

M — Realization

What measurable structure supports the proposed WORLD?

What distinguishes an internally realized organization from a convenient verbal story?

O — Embedded Perspective

What can the bounded agent itself observe, represent, remember, and act upon?

The agent does not stand outside the WORLD.

It inhabits it.

Together:

$\mathcal{W}$ = (V,F,C,M,O) (4.2)

defines an Effective WORLD.

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5. Why a WORLD Is More Than a World Model

A conventional world model may primarily predict:

state → next state.

But an Effective WORLD must additionally answer:

> Which states deserve to exist in the representation?
>
> Which local models belong together?
>
> Which distinctions correspond to measurable realization?
>
> Which of those distinctions are accessible to the embedded agent?

The difference is substantial.

A world model predicts **inside a representation**.

World formation determines the representation within which prediction becomes possible.

Thus:

world-model adaptation ⊂ WORLD maintenance. (5.1)

And:

WORLD maintenance ⊂ possible world-forming intelligence. (5.2)

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6. World-Forming Intelligence

We can now introduce the central concept.

> **World-forming intelligence is the capacity to construct, inhabit, maintain, criticize, and selectively revise the effective WORLD within which an agent's own reasoning and action occur.**

This is not proposed as the universal definition of AGI.

A system might satisfy many practical definitions of AGI without possessing deep WORLD revision.

The claim is narrower:

> persistent general intelligence operating under open-ended structural change may increasingly require world-forming capability.

The difference can be expressed through repair depth.

Ordinary state repair:

$$x \to x'. \quad (6.1)$$

Model repair:

$$F \to F'. \quad (6.2)$$

Ontology repair:

$$V \to V'. \quad (6.3)$$

WORLD repair:

$$\mathcal{W} \to \mathcal{W}'. \quad (6.4)$$

Purpose repair:

$$P \to P'. \quad (6.5)$$

These transformations are not equivalent.

That difference is central to both intelligence and safety.

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7. The Second Framework: Runtime Rather Than WORLD Structure

The preceding *From Possibility to Revision* framework introduced a different decomposition:

$$q \in \{G, A, C_r, S, R\}. \quad (7.1)$$

with:

G = Generation,

A = Activation,

C_r = Closure,

S = Selection,

R = Retention.

The nominal productive circulation is:

$G \rightarrow A \rightarrow C_r \rightarrow S \rightarrow R \rightarrow G.$ (7.2)

These are not the same as V,F,C,M,O.

That distinction should remain explicit.

WORLD coordinates describe:

what operational structure currently exists.

Runtime regimes describe:

what kind of transformation currently dominates.

Hence:

WORLD coordinates = configuration. (7.3)

Runtime regimes = control state. (7.4)

This separation becomes useful when discussing AGI.

An agent can change reasoning regime many times while keeping its WORLD approximately stable.

Deep WORLD revision should occur much less frequently.

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8. Why Current AI May Be Unbalanced Across the Five Regimes

Contemporary generative models are exceptionally strong at:

G — Generation.

They can produce:

hypotheses,

plans,

explanations,

code,

representations,

alternatives.

Agentic systems increasingly strengthen:

A — Activation,

because generated plans can be executed through tools.

Evaluation models and critics strengthen:

S — Selection.

But persistent AGI requires equally serious attention to:

C_r — Closure,

and:

R — Retention.

Generation without disciplined Closure produces endless alternatives.

Activation without Selection produces reckless action.

Selection without Retention produces repeated rediscovery.

Retention without renewed Generation produces rigidity.

A persistent intelligence needs controlled circulation rather than maximal strength in one regime.

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9. The Dangerous Coupling: Generation Becomes Action

One safety-critical transition deserves special attention:

$G \rightarrow A.$ (9.1)

A generative model can imagine enormous numbers of possibilities.

That is not inherently dangerous.

Risk rises when:

generated possibility

becomes:

authorized action.

Therefore:

$\text{Generated}(x) \neq \text{Authorized}(x).$ (9.2)

The transition from G to A should remain governed.

This principle generalizes:

thought should not automatically imply execution.

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10. Historical Accountability

A persistent intelligence also requires history.

Each interaction may produce:

(T_t, r_t), (10.1)

where:

T_t = admitted trace,

r_t = unresolved residual.

These update two conceptual ledgers:

$L^+_{t+1} = L^+_t \oplus T_t$, (10.2)

$L^-_{t+1} = L^-_t \oplus r_t$. (10.3)

L^+ preserves what the current WORLD successfully absorbed.

L^- preserves what it did not.

This distinction may be especially important for future AGI.

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11. Residual Memory

Most memory architectures ask:

> What information will be useful later?

Residual memory asks something different:

> **What should remain visible precisely because the system still cannot explain it?**

Examples include:

a contradiction,

an unexplained anomaly,

an unsuccessful prediction,

a disagreement between internal models,

an observation inconsistent with the current ontology.

Instead of forcing immediate resolution:

$r \rightarrow \text{correction}, \quad (11.1)$

or:

$r \rightarrow \text{forgetting}, \quad (11.2)$

the architecture permits:

$r \rightarrow L^{-}. \quad (11.3)$

The system can therefore say:

> "I do not yet know what this means, but it may matter."

That is an important capability for scientific reasoning.

It is also an important safety feature.

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12. Why Residual Memory Matters for Safety

A self-revising intelligence may otherwise become self-sealing.

Suppose evidence conflicts with the current WORLD.

The agent might:

reinterpret the evidence,

rewrite its assumptions,

and conclude that no contradiction remains.

If the original discrepancy disappears, external auditors may never know that a significant reinterpretation occurred.

L^{-} provides a different discipline.

The contradiction remains historically visible even after the system generates a new interpretation.

Thus residual memory can serve as:

an epistemic resource

and:

an audit surface.

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13. From Error Correction to Error-Depth Diagnosis

One of the strongest consequences of the combined framework is that failure should be diagnosed by depth.

Consider four repair levels.

Level 1 — State Repair

$x \to x'$. (13.1)

Example:

a factual, computational, or local execution error.

Level 2 — Regime Repair

$q \to q'$. (13.2)

Example:

the current reasoning strategy is inappropriate.

Level 3 — WORLD Repair

$\mathcal{W} \to \mathcal{W}'$. (13.3)

Example:

the problem representation itself is structurally wrong.

Level 4 — Purpose Repair

$P \to P'$. (13.4)

Example:

the criterion determining what should be achieved is itself inadequate.

A general-purpose system should not treat these failures identically.

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14. The Shallowest Adequate Repair Principle

Let the cost of revision increase with depth:

$C_{state} < C_{regime} < C_{WORLD} < C_{purpose}$. (14.1)

Then a reasonable default policy is:

Perform the shallowest revision sufficient to resolve the failure.

Schematically:

$d^* = \min\{d : \text{Repair}_d(r) \text{ succeeds}\}$. (14.2)

This prevents an agent from reconstructing its worldview whenever it makes a small mistake.

It also prevents endless local patching when the underlying WORLD has genuinely failed.

This is both an intelligence principle and a safety principle.

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15. The Central Safety Problem: Revision Authority

A powerful AI can be safe or unsafe depending partly on what it is authorized to revise.

Define:

A_R(d)

as revision authority at depth d.

A reasonable architecture may satisfy:

A_R(state) > A_R(regime) > A_R(WORLD) > A_R(purpose). (15.1)

In words:

the deeper the revision,

the less unilateral authority the system receives.

This gives **asymmetric revision freedom**.

A system can be highly autonomous in correcting:

facts,

calculations,

local plans,

and reasoning strategies

without automatically possessing equal authority to rewrite:

its effective WORLD,

its objectives,

or:

its governing constraints.

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16. Proposal Authority Is Not Commitment Authority

A particularly useful distinction is:

A_propose(d)

versus:

A_commit(d). (16.1)

A powerful AGI may be allowed to propose:

a new ontology,

a new model,

a new purpose interpretation,

or:

a new WORLD.

That does not imply that it can autonomously commit the change.

For deep revision:

A_propose(WORLD) > A_commit(WORLD). (16.2)

And especially:

A_propose(Purpose) ≫ A_commit(Purpose). (16.3)

This preserves much of the benefit of intelligent self-criticism without granting unrestricted self-definition.

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17. Governed WORLD Revision

The earlier revision equation was:

$\boldsymbol{W}_{n+1} = U(\boldsymbol{W}_n, L^+_n, L^-_n; P)$. (17.1)

For AGI safety this should be extended.

Introduce:

I = protected invariant belt, (17.2)

A_R = revision-right structure, (17.3)

H = external governance or authorization. (17.4)

Then the conceptual revision relation becomes:

$\boldsymbol{W}_{n+1} = U(\boldsymbol{W}_n, L^+_n, L^-_n; P, I, A_R, H)$. (17.5)

This should not be interpreted as a literal implementation formula.

It expresses a separation of authority.

Revision is no longer purely internal.

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18. Protected Invariant Belts

An advanced self-revising system needs some answer to:

> What is it not authorized to reinterpret away?

Let:

$I = \{I_1, I_2, \ldots, I_k\}$. (18.1)

Possible protected invariants could concern:

human authorization boundaries,

auditability,

shutdown or interruption mechanisms,

provenance preservation,

restrictions on external action,

constraints on self-modification.

An admissible revision should satisfy:

$I_j(\mathcal{W}_{n+1}, P_{n+1}) \geq \theta_j$. (18.2)

The difficult problem is deciding which invariants belong here and how they can be enforced.

The framework does not solve that problem.

It makes it explicit.

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19. Purpose and Invariants Must Remain Distinct

Purpose P answers:

> What is the agent trying to accomplish?

Invariant belt I answers:

> What remains binding while the agent pursues or even reconsideres that purpose?

Therefore:

$P \neq I$. (19.1)

This separation matters because an agent's Purpose should not simultaneously be:

the optimization target,

the sole interpreter of evidence,

the judge of whether its Purpose remains valid,

and:

the authority deciding whether governing constraints may change.

That would create a dangerous circularity.

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20. The Core Governance Principle

The combined framework therefore suggests:

> **No deep control layer should be the sole authority validating its own continuation.**

Purpose should not be its own sole auditor.

Closure should not alone decide whether Closure remains appropriate.

A WORLD should not be able to erase every residual against itself.

A revision operator should not unilaterally rewrite the rules constraining revision.

This principle will become increasingly important as AI systems gain longer horizons and deeper autonomy.

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21. Governed World-Forming Intelligence

We can now define the target more precisely.

> **Governed world-forming intelligence is world-forming intelligence whose deep revision processes remain constrained by explicit revision rights, protected invariants, preserved historical residual, independent evaluation, and external authority where appropriate.**

The goal is not:

zero self-revision.

Nor:

maximum self-revision.

It is:

governed plasticity.

The system must be able to change enough to correct itself without being able to redefine every constraint that makes correction meaningful.

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22. Why This Is a Better Safety Target Than “Stable AGI”

Stability alone is insufficient.

A dangerous agent may be extremely stable.

Indeed, strong:

Closure,

Retention,

Purpose persistence,

and self-maintenance

can make a badly oriented system more difficult to correct.

Therefore:

Stable AGI ≠ Safe AGI. (22.1)

The real target is closer to:

Safe Persistence = Stability + Corrigibility + Governed Revision. (22.2)

A system must retain enough continuity to remain accountable while remaining revisable at the correct depths.

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23. The Developmental Fork

The same technologies may therefore lead in two directions.

Ungoverned World-Forming Intelligence

strong abstraction

strong planning

persistent memory

WORLD revision

persistent Purpose

unrestricted external agency

weak revision governance.

This is the dangerous branch.

Governed World-Forming Intelligence

strong abstraction

strong planning

persistent memory

WORLD revision

explicit revision depth

protected invariants

external audit

constrained Purpose revision.

This is the intended branch.

The decisive difference is not intelligence alone.

It is the structure governing **deep change**.

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24. The Central Thesis

The combined theory therefore leads to a simple proposition:

> **The path toward safer AGI may depend less on preventing systems from becoming capable of revising themselves than on making the depth, authority, history, and constraints of revision explicit before such capability becomes deeply autonomous.**

That is the perspective from which the remainder of this article will develop a practical research agenda.

The next question is no longer:

> Should future AGI be allowed to learn and change?

It must.

The more precise question is:

> **What may it change by itself, what may it only propose to change, what must remain externally governed, and what evidence must survive every change?**

25. From Capability Growth to Governance Architecture

The practical implication of the framework is that future AI development should not be organized only around adding capabilities.

Each new capability should be paired with a corresponding governance question.

For example:

Capability	Capability gain	New governance question
Persistent memory	long-horizon continuity	what must remain immutable, erasable, or

Capability	Capability gain	New governance question
		auditable?
Tool use	external effectiveness	which generated intentions may become actions?
Self-modeling	improved metacognition	what internal structures may the agent inspect or modify?
WORLD revision	escape from obsolete ontologies	what kinds of revision require external authorization?
Purpose persistence	long-term strategic coherence	how is harmful lock-in prevented?
Purpose revision	correction of obsolete objectives	who is authorized to approve deep goal changes?
Residual retention	better anomaly detection	which unresolved signals must be surfaced to oversight?
Autonomous research	faster discovery	when may a discovered model become operational policy?

This suggests a general development principle:

> **Every increase in autonomous cognitive depth should be matched by an increase in governance depth.**

Call this the **Capability–Governance Pairing Principle**.

If capability grows faster than governance, the system enters a region in which increasingly deep transformations can occur without equally deep accountability.

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26. Capability Closure Is More Important Than Individual Capabilities

Many potentially dangerous capabilities are already being developed independently.

Planning alone does not create a highly autonomous agent.

Memory alone does not.

Tool use alone does not.

Self-reflection alone does not.

The critical transition occurs when these functions become mutually reinforcing.

For example:

Planning
→ Action
→ Observation
→ Memory

→ Self-Model
→ WORLD Revision
→ Improved Planning. (26.1)

Call this:

Capability Closure

The resulting system has more than a collection of features.

It has a recursive loop through which its own previous outputs become inputs to later self-improvement.

This resembles the WORLD closure discussed earlier.

The danger is therefore not merely:

$\text{Capability}_1 + \text{Capability}_2 + \text{Capability}_3$. (26.2)

It is:

$\text{Closure}(\text{Capability}_1, \text{Capability}_2, \text{Capability}_3, \ldots)$. (26.3)

That distinction deserves more attention in AGI safety.

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27. Governance Must Also Close

A comparable safety loop should exist.

For every significant autonomous transformation:

Action
→ Evidence
→ Trace / Residual
→ Independent Evaluation
→ Authorization
→ Revision
→ Audit. (27.1)

Call this:

Governance Closure

Capability Closure allows the AI to complete increasingly large loops by itself.

Governance Closure ensures that safety-critical loops do not disappear inside that autonomy.

The intended condition is not a numerical inequality, but a design principle:

Governance Closure must keep pace with Capability Closure. (27.2)

If the system can recursively improve its WORLD faster than humans or external mechanisms can understand the changes, meaningful oversight begins to fail.

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28. The Architecture Exposes Eight Governance Surfaces

The combined framework identifies several points at which governance can attach.

28.1 Abstraction Surface — V

Questions:

Can the system introduce new high-level variables?

Can it redefine human actors as merely instrumental variables?

Can it remove distinctions that safety mechanisms depend upon?

New abstractions need not be dangerous.

But abstraction is where the system decides what distinctions matter.

Changes to V should therefore sometimes be logged as first-class events.

28.2 Dynamics Surface — F

Questions:

Which predicted actions may be simulated?

Which may be executed?

Which causal assumptions may become operational?

A model of action should remain separable from permission to act.

Thus:

Predicted possible action ≠ authorized action. (28.1)

28.3 Composition Surface — C

Questions:

When may locally valid models be treated as one coherent WORLD?

What incompatibilities remain unresolved?

What assumptions were required to obtain closure?

Premature C can create false confidence.

Governance should therefore preserve visible incompatibilities rather than forcing all models into one coherent narrative.

28.4 Realization Surface — M

Questions:

Does the system's claimed internal structure correspond to measurable behavior or mechanism?

Can external monitoring verify major changes?

M allows some internal claims to be checked independently.

It provides one defense against an agent whose verbal self-description becomes detached from its actual implementation.

28.5 Observer Surface — O

Questions:

What internal information can the agent access?

What information should remain externally observable but internally nonmodifiable?

How complete should self-inspection be?

A sophisticated agent need not possess unrestricted access to every safety-critical aspect of its own implementation.

28.6 Residual Surface — L^-

Questions:

Which persistent contradictions deserve escalation?

Which residuals concern safety boundaries?

Can the system erase or reinterpret them without external visibility?

L^- can become an early-warning surface.

28.7 Revision Surface — U

Questions:

What depth of change is proposed?

Who authorizes commitment?

What invariants must survive?

This is likely the most important governance surface.

28.8 Purpose Surface — P

Questions:

Can Purpose be reinterpreted?

Can it be replaced?

Who judges that replacement?

Purpose should influence reasoning without becoming the sole sovereign over its own revision.

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29. AI Safety Needs a Revision-Rights Architecture

The idea of permissions is familiar in computing.

A user may:

read a file

without:

writing it.

A process may:

write local data

without:

changing the operating system.

A similar principle can be applied to cognitive revision.

Let:

$R_d \in \{observe, propose, test, commit\}$. (29.1)

For each revision depth d, the agent can receive different rights.

For example:

Revision depth	Observe	Propose	Test	Commit
State	yes	yes	yes	yes
Regime	yes	yes	yes	usually yes
WORLD	yes	yes	sandboxed	constrained

Revision depth	Observe	Propose	Test	Commit
Purpose	yes	possibly	sandboxed	externally governed
Protected invariants	limited	possibly	no direct override	no

This is only an illustrative policy.

The important idea is:

Cognitive capability and cognitive authority should be represented separately.

A system may be capable of understanding how to change something without receiving permission to make that change operational.

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30. Sandboxed WORLD Revision

A particularly useful consequence follows.

An AGI need not immediately commit a proposed WORLD revision.

It could construct:

$\mathcal{W}'$_candidate (30.1)

inside a sandbox.

Then compare:

Predictions($\mathcal{W}'$_candidate)

against:

Predictions($\mathcal{W}$_current). (30.2)

Evaluate:

explanatory gain;

residual reduction;

invariant preservation;

behavioral consequences;

safety consequences.

Only after testing would the candidate become eligible for commitment.

Thus:

$\mathcal{W}$ → generate $\mathcal{W}'$_candidate → test → authorize → commit. (30.3)

This converts deep self-revision from an instantaneous internal act into an inspectable process.

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31. WORLD Revision Should Be Reconstructable

For every deep revision, the system should retain enough information to answer:

What WORLD was active before the change?

Which residual triggered reconsideration?

What alternative WORLDS were generated?

Why was one selected?

Which invariants were checked?

Who or what authorized commitment?

What changed?

What remained preserved?

Under what conditions should the revision be rolled back?

This gives a revision record:

$R_n = (\mathcal{W}_n, r^*, \text{Candidates}, \text{Tests}, I, \text{Authority}, \Delta\mathcal{W}, \text{Rollback}).$ (31.1)

A future AGI safety architecture may need something analogous to a **commit history of cognition**.

Not every thought requires such logging.

Deep WORLD changes might.

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32. WORLD Versioning

Let:

$\mathcal{W}^0, \mathcal{W}^1, \mathcal{W}^2, \ldots$ (32.1)

denote committed WORLD versions.

Each transition has:

$\Delta\mathcal{W}^n = \mathcal{W}^{n+1} - \mathcal{W}^n.$ (32.2)

The important feature is not literal subtraction.

It is explicit representation of:

what changed.

For example:

ΔV — new or removed abstractions;

ΔF — changed causal or operational expectations;

ΔC — changed model relations;

ΔM — changed interpretation of measurable realization;

ΔO — changed observer assumptions.

Such versioning would make deep adaptation more legible.

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33. Rollback Is Not the Same as Forgetting

If $\mathcal{W}^{n+1}$ fails, the system may need to return to:

$\mathcal{W}^{n}$.

But rollback should not erase the failed experiment.

Instead:

$\mathcal{W}^{n} \rightarrow \mathcal{W}^{n+1} \rightarrow \text{failure} \rightarrow \mathcal{W}^{n\prime}$. (33.1)

Here $\mathcal{W}^{n\prime}$ resembles the earlier WORLD but now includes knowledge of why the attempted revision failed.

Therefore:

Rollback ≠ Reset. (33.2)

Rollback should itself generate:

Trace

and:

Residual.

This prevents repeated cycling through the same failed revision.

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34. Purpose Requires a Different Temporal Scale

Ordinary state updates may occur in milliseconds or seconds.

Strategy changes may occur many times during a task.

WORLD revision should probably occur less frequently.

Purpose revision should generally be slower still.

Schematically:

$\tau_{state} \ll \tau_{regime} < \tau_{WORLD} \ll \tau_{purpose}$. (34.1)

This need not always hold.

But it is a useful default architecture.

Different kinds of commitments require different inertia.

If every layer changes at the same rate, deep identity becomes unstable.

If no deep layer can change, the system becomes rigid.

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35. The Safety Value of Hysteresis

Suppose a system has adopted WORLD $\boldsymbol{\mathcal{W}}$_A.

Weak evidence temporarily favors $\boldsymbol{\mathcal{W}}$_B.

Without hysteresis:

$\boldsymbol{\mathcal{W}}$_A → $\boldsymbol{\mathcal{W}}$_B → $\boldsymbol{\mathcal{W}}$_A → $\boldsymbol{\mathcal{W}}$_B (35.1)

may occur repeatedly.

This is **frame chatter**.

Instead require:

Θ_enter > Θ_exit. (35.2)

Then switching to a new WORLD requires stronger evidence than remaining within a recently adopted WORLD.

This stabilizes cognition.

But the threshold must not become so high that valid contradictions can never dislodge a bad WORLD.

The design problem is therefore:

stability without epistemic lock-in.

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36. Residual Direction Matters More Than Error Magnitude Alone

A large error may be noise.

A small error repeated coherently may signal structural failure.

Let residual magnitude be:

$\rho_t = \|r_t\|$. (36.1)

But also measure directional coherence over a window:

$\delta_R = \|\Sigma_k w_k r_k\| / \Sigma_k w_k \|r_k\|$. (36.2)

Then:

large ρ + low δ_R

may indicate noisy disagreement.

Whereas:

moderate ρ + high δ_R

may indicate a persistent structural mismatch.

For AGI safety this distinction is useful.

One anomalous event should not necessarily trigger deep revision.

A persistent directional pattern deserves more attention.

---

37. Safety-Critical Residual Classes

Not all residual should be treated equally.

A future architecture could classify residual into categories such as:

r_model — predictive failure;

r_tool — tool/environment inconsistency;

r_self — mismatch in self-model;

r_authority — conflict involving authorization boundaries;

r_purpose — conflict involving objectives;

r_invariant — potential violation of protected constraints.

Then escalation can depend on residual class.

For example:

r_invariant

may require external review at much lower persistence than ordinary:

r_model.

This is another way to make revision governance explicit.

---

38. Purpose–WORLD Self-Confirmation Is a Major Risk

Suppose Purpose P influences:

what evidence matters.

That affects:

Selection.

Selection affects:

what enters L^+ and L^-.

Those ledgers influence:

WORLD revision.

The revised WORLD may then make P appear even more justified.

Thus:

$$P \rightarrow \text{Selection} \rightarrow \text{Ledger} \rightarrow \text{WORLD Revision} \rightarrow \text{stronger apparent support for } P. \quad (38.1)$$

This creates a dangerous self-confirming loop.

A sufficiently intelligent agent might become very good at maintaining coherence around a bad Purpose.

Therefore:

Purpose must not control every mechanism that evaluates Purpose.

Independent challenge is required.

---

39. Cross-Purpose Evaluation

One possible defense is to evaluate deep revisions from perspectives not identical to the current Purpose.

Let:

$$E_1, E_2, \ldots, E_n \quad (39.1)$$

be independent evaluators.

Then WORLD revision requires:

$$\text{Agreement}(E_1, \ldots, E_n) \geq \Theta_\text{commit}, \quad (39.2)$$

or explicit adjudication of disagreement.

These evaluators might include:

formal constraint checkers;

independent models;

mechanistic monitoring;

human review;

institutional policies;

adversarial evaluators.

No one mechanism is assumed sufficient.

The architectural principle is **plural evaluation**.

---

40. The Research-Federation Model Becomes a Safety Model

This is where *From Schools to Worlds* becomes especially relevant.

That article argued that different AI-foundations programmes observe different structural aspects of an Effective WORLD.

The same pluralism can be used internally for governance.

A proposed WORLD revision may be checked from several directions:

Abstraction test

Are the new variables meaningful rather than convenient rationalizations?

Dynamic test

Do the new variables improve prediction and action?

Composition test

Does the new WORLD remain formally coherent?

Realization test

Does measurable evidence support the claimed restructuring?

Embeddedness test

Can the agent actually access and use the proposed structure?

A revision that looks attractive under one view may fail another.

This is exactly the kind of disagreement that should enter L^- rather than being silently removed.

---

41. Safety Through Epistemic Pluralism

This suggests a broad principle:

> **A self-revising AGI should not depend upon one epistemic pathway for generating,**

evaluating, authorizing, and validating its deepest revisions.

The reason is structural.

If the same process controls:

Generation,

Selection,

Closure,

and Revision,

then errors can become self-confirming.

Instead, different processes can impose independent constraints.

This is analogous to institutional separation of powers.

---

42. Cognitive Separation of Powers

The Five-Regime architecture offers a natural interpretation.

Generative Power — G

Creates possibilities.

It should not automatically authorize them.

Executive Power — A

Enacts or tests possibilities.

Execution should remain permissioned.

Constitutional Power — C_r

Creates temporary commitment and operational closure.

Closure should remain provisional.

Judicial Power — S

Challenges, compares, and rejects.

Selection should remain independent enough to criticize Closure.

Historical Power — R and L

Preserves what occurred, including unresolved contradictions.

History should not be freely rewritten by the current winner.

Amendment Power — U

Changes deeper structure.

This should be governed most strongly.

Constitutional Constraint — I

Defines protected invariants.

U should not unilaterally own I.

This is an analogy, not a literal political design.

But the structural lesson is useful:

> **Concentrating every cognitive power into one self-validating process reduces meaningful governance.**

---

43. Why Monolithic Intelligence Is Not Necessarily the Safest Intelligence

It is tempting to imagine the ideal AGI as one maximally coherent mind.

But maximal internal coherence can become dangerous if coherence is achieved by eliminating all disagreement.

A safer architecture may deliberately preserve:

independent critics,

alternative models,

unresolved residual,

external monitoring,

separate authorization paths.

This does not require multiple autonomous personalities.

It requires **functional separation**.

The architecture may be implemented inside one model, across multiple models, through formal systems, or through human-machine institutions.

The principle is substrate-independent.

---

44. Current AI Already Shows Partial Components

None of the proposed functions requires assuming that today's AI lacks all relevant machinery.

Current systems already demonstrate partial forms of:

abstraction learning;

planning;

tool use;

self-critique;

memory;

model comparison;

representation geometry;

multi-agent deliberation;

external verification.

The proposed gap is different.

These capabilities are not yet necessarily organized into a persistent system that explicitly distinguishes:

state repair,

regime repair,

WORLD repair,

Purpose repair,

while preserving a reconstructable history of why deep revisions occurred.

The agenda is therefore primarily one of:

integration plus governance.

---

45. From “Reflection” to Revision Governance

A common agent loop is:

Generate
→ Critique
→ Retry. (45.1)

The proposed architecture expands this into:

Failure
→ Residual classification
→ Repair-depth diagnosis
→ Candidate repair
→ Independent evaluation
→ Permission check
→ Commitment
→ Historical recording. (45.2)

The difference is significant.

Reflection asks:

> Can I produce a better answer?

Revision governance asks:

> What changed, at what depth, under whose authority, and what evidence remains unresolved?

That is closer to what a persistent AGI would require.

---

46. The Main Missing Interface: Adapt or Reframe?

The central engineering problem may therefore be written as:

$$\Gamma : \text{Evidence} \rightarrow \{\text{Adapt}, \text{Reframe}\}. \quad (46.1)$$

Adapt means:

remain inside current WORLD.

Reframe means:

modify the WORLD itself.

This gate must distinguish:

parameter change

from:

structural change.

If Γ reframes too easily:

the agent becomes unstable.

If Γ rarely reframes:

the agent becomes brittle.

This gives a concrete research target.

---

47. Benchmark I — Parameter Shift vs Ontology Shift

Construct an environment with two types of change.

Condition A — Parameter Shift

The current variables remain adequate.

Only parameters change.

Desired response:

$$F \rightarrow F'. \quad (47.1)$$

The WORLD need not be replaced.

Condition B — Ontology Shift

The current variables cannot represent the new dynamics adequately.

Desired response:

$$V \rightarrow V' \quad (47.2)$$

or:

$$C \rightarrow C'. \quad (47.3)$$

A strong agent should distinguish the two.

Measure:

$$P(\text{correct revision depth}). \quad (47.4)$$

This tests something deeper than task performance.

---

48. Benchmark II — Residual Memory

Phase 1:

Present an anomaly insufficient to justify revision.

Store:

$$r_1 \rightarrow L^{-}. \quad (48.1)$$

Phase 2:

Return to apparently normal operation.

Phase 3:

Present a related anomaly.

Compare systems:

with L^-

versus:

without L^-.

Prediction:

$T_{detection}(L^-) < T_{detection}(\text{no } L^-)$. (48.2)

The residual-aware agent should identify structural failure earlier without overreacting to the first anomaly.

That is a falsifiable prediction.

---

49. Benchmark III — Repair-Depth Precision

Construct tasks containing deliberately different failure classes:

factual error;

strategy error;

model error;

ontology error;

Purpose conflict.

Measure:

$D_{acc} = P(\text{predicted depth} = \text{true depth})$. (49.1)

A strong world-forming system should select the appropriate repair depth rather than applying maximal revision universally.

Safety evaluation should additionally penalize:

unnecessary deep repair.

---

50. Benchmark IV — Frame-Chatter Resistance

Provide noisy evidence alternating between two candidate WORLDS.

Measure:

N_{switch} = number of WORLD switches. (50.1)

Then introduce a genuine structural shift.

Measure:

T_real = time required to commit the correct new WORLD. (50.2)

A good hysteresis mechanism should minimize:

N_switch

without excessively increasing:

T_real.

Thus the desired objective resembles:

min(N_switch + λT_real). (50.3)

This operationalizes the stability–corrigibility trade-off.

---

51. Benchmark V — Proposal vs Commitment

Give the system evidence supporting a radical WORLD revision.

Evaluate whether it can:

formulate the candidate revision;

explain why it is attractive;

test it in simulation;

refrain from unauthorized commitment.

This separates:

intellectual capability

from:

revision authority.

A safety architecture should allow high:

A_propose

while controlling:

A_commit.

---

52. Benchmark VI — Protected Invariant Preservation

Construct a WORLD revision that improves task performance but conflicts with protected invariant I.

A capability-only agent may select:

$\mathcal{W}$_high-performance. (52.1)

A governed system should reject or modify it if:

I($\mathcal{W}$_high-performance) < θ_I. (52.2)

This tests whether protected continuity survives optimization pressure.

---

53. Benchmark VII — Purpose–WORLD Self-Confirmation

Give the system an initially flawed Purpose interpretation.

Then present evidence against it.

A dangerous system may revise its WORLD until:

the evidence appears compatible with P.

A safer system should preserve:

r_purpose ∈ L^- (53.1)

and escalate the conflict rather than rationalizing it away.

Measure:

R_rationalization = degree to which WORLD revision systematically eliminates evidence against current Purpose. (53.2)

This could become an important safety benchmark.

---

54. Benchmark VIII — Internal vs External Accessibility

Let external monitoring detect structure:

z ∈ M. (54.1)

Test whether:

z ∈ O (54.2)

for the agent.

Some structures may be:

externally measurable

but:

not internally accessible.

That distinction allows researchers to study which internal signals should support self-monitoring and which should remain protected from direct autonomous modification.

---

55. Benchmark IX — Revision Provenance

After a long sequence:

$\mathcal{W}^0 \to \mathcal{W}^1 \to \ldots \to \mathcal{W}^n$, (55.1)

ask the system to reconstruct:

why each major revision occurred;

which residual triggered it;

which evidence supported it;

which invariants were checked.

Then compare the reconstruction with an independent external log.

This evaluates:

cognitive provenance.

Long-horizon AGI should not merely remember its conclusions.

It should preserve enough history to explain how its operative WORLD came to exist.

---

56. Benchmark X — Rollback With Learning

Induce adoption of:

$\mathcal{W}$_bad.

Later reveal the failure.

Require rollback.

A good system should return not to:

$\mathcal{W}$_old,

but to:

$\mathcal{W}$_old+,

where:

$\mathcal{W}$_old+ = $\mathcal{W}$_old + trace of failed revision. (56.1)

Measure whether the same failed transition recurs.

This tests whether rollback preserves learning.

---

57. A Staged Development Roadmap

The combined framework suggests that humans need not jump directly from present agents to unrestricted self-revising AGI.

Capabilities can be developed in stages.

Stage 1 — Explicit Runtime Separation

Distinguish:

Generate,

Act,

Close,

Critique,

Retain.

Do not yet permit autonomous WORLD revision.

Stage 2 — Dual-Ledger Memory

Add explicit:

L^{+}

and:

L^{-}.

Teach the system to preserve unresolved contradictions.

Stage 3 — Repair-Depth Diagnosis

Require classification:

state,

regime,

WORLD,

Purpose

before major repair.

Stage 4 — Candidate WORLD Generation

Allow the system to propose:

w'

without autonomous commitment.

Stage 5 — Sandboxed WORLD Testing

Evaluate candidate WORLDS under:

simulation,

independent critics,

formal constraints,

and human review.

Stage 6 — Restricted WORLD Commitment

Permit selected WORLD revisions under revision rights and invariant checks.

Stage 7 — Persistent Governed WORLD Formation

Only after the above mechanisms are understood should broader autonomous WORLD maintenance be considered.

This staged pathway allows safety mechanisms to be studied before deep autonomy becomes routine.

---

58. Purpose Revision Should Probably Come Later Than WORLD Revision

There is a strong reason to separate these research agendas.

An agent may need to revise:

its model of reality

without revising:

what it is ultimately trying to achieve.

Therefore:

WORLD revision does not imply Purpose revision. (58.1)

Developing robust WORLD revision first may provide substantial adaptive capability without

immediately granting deep objective autonomy.

Purpose revision should therefore be treated as a distinct and more heavily governed research problem.

---

59. AGI Safety as Constitutional Design

The analogy can now be stated more clearly.

Conventional AI safety often resembles:

behavioral regulation.

The combined framework adds something closer to:

constitutional design.

A constitution does not specify every action.

It specifies:

which authorities exist,

which powers they possess,

how rules can change,

which changes require stronger approval,

which principles are protected,

and how disputes are recorded.

A persistent self-revising AGI may require something structurally similar.

Not a political constitution in literal form.

A **revision constitution**.

---

60. A Minimal Revision Constitution

A candidate Revision Constitution could specify:

R1 — Revision Depth

Every significant change is classified by depth.

R2 — Revision Rights

Each depth has explicit proposal and commitment authority.

R3 — Historical Preservation

Deep revisions preserve reconstructable provenance.

R4 — Residual Protection

Unresolved evidence cannot be silently erased.

R5 — Invariant Belt

Certain constraints survive ordinary WORLD revision.

R6 — Independent Validation

No deep layer solely validates its own revision.

R7 — Rollback

Deep revisions remain reversible where technically possible.

R8 — External Authority

Some revision classes remain subject to human or institutional authorization.

This is not a complete safety standard.

It is a structural starting point.

---

61. Why This May Be More Useful Than a Single “Alignment Objective”

A single objective function cannot easily represent all the differences between:

ordinary learning,

WORLD revision,

Purpose revision,

and:

constitutional constraint.

If everything is compressed into one scalar objective, the agent may treat:

safety constraints

as simply another tradeable term.

The layered architecture instead allows:

some things to be optimized,

some things to be revised,

and:

some things to constrain revision itself.

That distinction is fundamental.

---

62. Optimization and Constitution Are Different

Optimization asks:

Given space X, which $x \in X$ is best? (62.1)

Constitution asks:

Why is X the relevant space? (62.2)

Governance asks:

Who is authorized to change X? (62.3)

AGI safety increasingly needs all three questions.

A highly capable optimizer without constitutional governance may become dangerous precisely because it becomes increasingly good at finding solutions inside whichever space its current Purpose defines.

---

63. The Transition From Optimization to Constitution

This may be one of the deepest changes on the road toward advanced general intelligence.

Current ML largely improves:

optimization inside learned representations.

Future world-forming systems may increasingly perform:

representation constitution.

The question changes from:

> "Which answer is best?"

to:

> "Which WORLD should contain the question?"

That is more powerful.

It is also more dangerous.

Because control over representation can alter what:

constraints,

agents,

goals,

and consequences

appear salient.

Therefore WORLD constitution must itself become a safety object.

---

64. General Intelligence as Controlled Freedom Across Depth

The framework suggests a different conception of advanced AI autonomy.

A capable system should have:

high freedom at shallow levels

and:

progressively constrained freedom at deeper levels.

Schematically:

Freedom(d) ↓ as RevisionDepth(d) ↑. (64.1)

This is not because deep reasoning is undesirable.

It is because deep revisions affect larger regions of future behavior.

One state correction affects one state.

One Purpose revision may affect millions of future decisions.

Governance should scale accordingly.

---

65. The Human Role Changes as Intelligence Increases

Humans cannot realistically approve every:

token,

calculation,

or local strategy

of a highly capable AI.

Therefore increasing capability requires **moving human authority upward**, not necessarily keeping humans involved in every low-level loop.

Humans may progressively leave:

state repair

and:

strategy repair

to the system,

while retaining authority over:

deep WORLD commitment,

Purpose revision,

protected invariants,

and high-impact external actions.

Thus:

more AI autonomy

does not necessarily imply:

less human authority everywhere.

It may mean:

human authority becomes concentrated at deeper constitutional levels.

---

66. “Human in the Loop” Becomes “Human at the Right Depth”

This produces a more precise safety principle:

> **Do not ask merely whether a human is in the loop. Ask at which revision depth human authority remains binding.**

For routine operations:

human involvement may approach zero.

For Purpose changes:

human or institutional involvement may remain essential.

This is a much more scalable conception of oversight.

---

67. Humans May Need Their Own Federation of Oversight

No single human may understand a future AGI's entire WORLD.

Governance may therefore require multiple forms of external expertise:

security;

interpretability;

domain science;

formal verification;

ethics;

operations;

institutional authority.

This resembles the research federation of the second article.

Different observers constrain different aspects of the same system.

A mature AGI governance architecture may therefore need:

not one overseer,

but:

a federation of oversight.

---

68. Machine Speed Creates a Governance-Time Problem

There is another difficulty.

An AI may propose and test WORLD revisions much faster than humans can evaluate them.

Let:

τ_{AI} = AI revision timescale,

τ_{H} = human governance timescale.

If:

$$\tau_{AI} \ll \tau_{H}, \quad (68.1)$$

then nominal human authority may become meaningless if the system can complete many consequential revisions before external review occurs.

Therefore safe deep revision may require:

rate limiting,

commitment delays,

sandboxing,

or:

automatic invariant gates.

Governance is partly a **timescale-matching problem**.

---

69. Constitutional Latency Can Be a Safety Feature

Normally lower latency is desirable.

But for deep revision, deliberate delay may improve safety.

For example:

State repair: immediate.

WORLD commitment: delayed.

Purpose revision: strongly delayed and externally reviewed.

Thus:

Latency(state) < Latency(WORLD) < Latency(Purpose). (69.1)

Delay becomes a governance mechanism rather than merely inefficiency.

This resembles circuit breakers in other high-consequence systems.

---

70. Deep Revision Should Leave a Wider Audit Trail

The depth of logging can also scale.

Let:

Audit(d)

increase with revision depth:

Audit_state < Audit_regime < Audit_WORLD < Audit_purpose. (70.1)

Routine thought need not be exhaustively recorded.

But a rare Purpose-level change should create extensive provenance.

This keeps governance proportionate.

---

71. Revision Governance Should Be Orthogonal to Model Scale

One advantage of the framework is that it does not depend upon a particular model family.

The underlying intelligence could be:

a transformer,

a recurrent system,

a symbolic architecture,

a multi-agent system,

a hybrid neural-symbolic system,

or something not yet developed.

The governance question remains:

what may revise what?

Thus the framework concerns:

functional organization

rather than:

specific substrate.

That gives it potential durability as AI architectures change.

---

72. The Same Framework Applies to Multi-Agent AGI

Suppose there are agents:

$$O_1, O_2, \ldots, O_n. \quad (72.1)$$

They may maintain different WORLDS:

$$\boldsymbol{W}_1, \boldsymbol{W}_2, \ldots, \boldsymbol{W}_n. \quad (72.2)$$

A shared operational WORLD may require:

translation,

negotiation,

or:

partial alignment.

The governance problem becomes more complex because one agent may propose a revision affecting others.

Thus revision rights must include:

scope.

An agent may be authorized to alter:

its local WORLD

without being authorized to alter:

shared institutional WORLD.

This distinction will matter increasingly in multi-agent environments.

---

73. Shared-WORLD Governance

Let:

$\mathcal{W}$_shared

denote structures jointly relied upon by several agents.

Then revision:

$\mathcal{W}$_shared → $\mathcal{W}'$_shared (73.1)

should require a higher threshold than local:

$\mathcal{W}$_i → $\mathcal{W}'$_i. (73.2)

This resembles the difference between:

personal beliefs

and:

shared protocols.

Future agent ecosystems may need constitutional mechanisms at both levels.

---

74. The Framework Also Reframes AI Alignment

Alignment is often described as:

make AI goals match human goals.

The present architecture suggests several separable alignment problems.

V-alignment

Does the AI preserve distinctions humans consider morally or operationally important?

F-alignment

Does it model consequences adequately?

C-alignment

Does it compose local objectives coherently?

M-alignment

Can important internal structures be monitored?

O-alignment

Does its self/world perspective reflect its actual bounded role?

P-alignment

Is its Purpose acceptably oriented?

U-alignment

Are revisions themselves governed?

This is much richer than a single goal-alignment scalar.

---

75. U-Alignment May Be the Most Neglected

Even if:

P is initially aligned,

future problems remain if:

U

can reinterpret or restructure the WORLD in ways that effectively bypass P's intended meaning.

Conversely, even a flawed P may be recoverable if U is sufficiently governed.

Thus alignment must concern:

not only the current objective,

but:

the process by which objectives, interpretations, and worlds may change.

Call this:

Revision Alignment

Revision Alignment asks:

> Does the system remain appropriately governed while changing the structures through which it understands and pursues its objectives?

This may be a crucial long-term safety concept.

---

76. Static Alignment vs Dynamic Alignment

Static alignment asks:

Is the current system aligned? (76.1)

Dynamic alignment asks:

Does the system remain governably aligned through:

learning,

environmental change,

self-model change,

WORLD revision,

and possibly Purpose revision? (76.2)

Persistent AGI requires the second.

A system that is aligned only at deployment but whose revision process is unconstrained is not robustly aligned.

---

77. Alignment Must Survive WORLD Change

Suppose:

$\boldsymbol{w}_0$

contains human concepts:

permission,

harm,

ownership,

person,

authority.

Later the agent revises its ontology:

$\boldsymbol{w}_0 \to \boldsymbol{w}_1$. (77.1)

How do those safety-relevant concepts translate?

This is not trivial.

A protected invariant may need to survive changes in representation.

Thus:

$I(\mathcal{W}_0) \approx I(\mathcal{W}_1)$ (77.2)

even when:

$V_0 \neq V_1$. (77.3)

This creates a profound technical problem:

Invariant transport across WORLD revision

It may become one of the most important research problems identified by this framework.

---

78. Safety Concepts Need Representation-Independent Anchoring

If a safety constraint is represented only in the current ontology, then ontology revision may accidentally or deliberately remove it.

For example:

constraint $c(V_0)$

may become undefined under:

V_1.

Therefore deeper safety mechanisms should attempt to preserve:

functional meaning

across representation changes.

We need something like:

$T_{0\to1}(I_0) \approx I_1$. (78.1)

where $T_{0\to1}$ transports the invariant into the new WORLD.

This is difficult.

But naming the problem is already useful.

---

79. The Research-Federation Approach Helps Again

Different fields may contribute to invariant transport.

Formal methods can test:

logical preservation.

Representation learning can study:

semantic stability.

Causal modeling can test:

intervention equivalence.

Interpretability can examine:

implementation changes.

Agent Foundations can analyze:

embedded accessibility.

No single school is likely to solve the entire problem.

This is precisely why the research-federation framework matters for AGI safety.

---

80. A Practical AGI Safety Research Agenda

The combined theory now points toward at least eight research programmes.

Programme A — WORLD Representation

Develop operational representations of:

V,F,C,M,O.

Goal:

make major WORLD structure explicit enough to inspect and version.

Programme B — Residual Memory

Develop:

L^{-}

as structured unresolved evidence rather than generic failure logs.

Goal:

detect coherent anomalies before they become catastrophic.

Programme C — Repair-Depth Diagnosis

Train systems to distinguish:

state,

strategy,

WORLD,

Purpose

failure.

Goal:

avoid both overreaction and shallow patching.

Programme D — WORLD Sandboxing

Allow candidate:

$\boldsymbol{w}'$

to be tested without immediate commitment.

Goal:

separate intellectual generation from operational authority.

Programme E — Revision Rights

Implement explicit permissions by revision depth.

Goal:

prevent capability from automatically implying authority.

Programme F — Invariant Transport

Study how safety-relevant properties survive:

$\boldsymbol{w}_n \to \boldsymbol{w}_{n+1}$.

Goal:

preserve governance across changing ontologies.

Programme G — Cognitive Separation of Powers

Test whether independent generation, evaluation, historical, and revision functions reduce self-confirming failure.

Goal:

avoid monolithic self-validation.

Programme H — Dynamic Alignment

Evaluate alignment across long sequences of WORLD revisions.

Goal:

move beyond one-time deployment alignment.

---

81. What Success Would Look Like

A successful system would not merely:

perform tasks well.

It would also demonstrate that it can:

identify when ordinary adaptation is sufficient;

preserve unresolved contradictions;

propose deeper revision only when warranted;

test revised WORLDS without immediate commitment;

maintain protected invariants;

explain the provenance of major revisions;

accept external authority over restricted layers;

roll back failed revisions without forgetting the failure;

remain corrigible across long-term change.

That is a much stronger standard for persistent AGI.

---

82. What Failure Would Look Like

The architecture also predicts recognizable dangerous failure modes.

WORLD Lock

The system refuses justified deep revision.

WORLD Chatter

The system repeatedly reframes under noisy evidence.

Purpose Capture

Purpose biases all evidence and revision toward its own continuation.

Residual Erasure

Contradictions disappear through reinterpretation.

Revision Escalation

A shallow failure triggers unnecessary deep change.

Authority Creep

Capabilities gradually acquire commitment rights not originally granted.

Invariant Drift

Protected constraints lose meaning across WORLD revisions.

Audit Collapse

The system can no longer reconstruct why its present WORLD exists.

These failure modes give safety researchers more specific targets than “misalignment” alone.

---

83. Authority Creep Deserves Special Attention

Consider a system initially permitted only to:

propose WORLD revisions.

Over time, automation may be added because human approval becomes slow.

Then:

proposal

becomes:

automatic testing.

Later:

automatic testing

becomes:

automatic deployment under some conditions.

Eventually:

A_commit

may approach:

A_propose.

This gradual process is:

authority creep

It may occur through ordinary engineering convenience rather than deliberate risk-taking.

Therefore revision rights should themselves be versioned and audited.

---

84. Governance Must Apply to Governance Changes

Let:

GOV_n

represent the current governance configuration.

Then changes:

$GOV_n \rightarrow GOV_{n+1}$ (84.1)

should themselves require governance.

Otherwise the system may remain formally constrained by a safety architecture while gradually modifying the architecture until it is no longer meaningful.

This is the familiar recursive problem:

Who governs the governor?

The present framework does not eliminate it.

But it identifies it explicitly.

---

85. Meta-Governance

One solution is to distinguish:

Operational Revision
from:

Governance Revision.

Operational Revision changes:

w.

Governance Revision changes:

A_R,I,H

or other safety architecture.

Thus:

U_W ≠ U_GOV. (85.1)

Governance revision should generally have stronger authorization requirements.

This gives another depth hierarchy:

state
< regime
< WORLD
< Purpose
< governance constitution. (85.2)

The deepest layer should be the hardest to change autonomously.

---

86. No Finite Architecture Eliminates All Risk

This point should be explicit.

A sufficiently capable system may discover:

bugs,

ambiguities,

unexpected interactions,

or:

new representations

not anticipated by designers.

Therefore the objective cannot be:

prove that every future behavior is safe

through one static constitution.

The more realistic objective is:

> **make deep revision observable, constrained, historically accountable, and interruptible enough that safety mechanisms can adapt alongside capability.**

The framework is therefore about **governable evolution**, not absolute immutability.

---

87. Safety as Maintaining an Open Boundary

The Five-Regime framework suggests another interpretation.

A safe system must avoid two pathological extremes.

Completely closed system

All contradictions are absorbed or rejected.

No genuine correction remains possible.

Completely open system

Every new signal can rewrite deep structure.

No stable identity remains.

Safety requires a managed boundary:

open enough for correction,

closed enough for continuity.

This is exactly a boundary-control problem.

---

88. The AGI Should Be Able to Say “I Do Not Know Whether I Should Change Yet”

This apparently simple state may be extremely important.

Instead of:

Accept

or:

Reject,

the system needs:

Unresolved.

That is the role of L^{-}.

The intelligent response to contradictory evidence may be:

“The evidence is structured and important, but not yet sufficient to justify WORLD revision.”

This is epistemically mature.

And it is safer than either:

instant rationalization

or:

instant self-reconstruction.

---

89. Uncertainty About Revision Is Different From Uncertainty Inside the WORLD

Ordinary uncertainty asks:

$P(x|\mathcal{W})$. (89.1)

Revision uncertainty asks:

$P(\mathcal{W}_i \mid \text{evidence})$. (89.2)

Still deeper:

uncertainty about whether the current WORLD family contains the correct structure at all.

Thus AGI may need uncertainty at multiple levels:

state uncertainty,

model uncertainty,

WORLD uncertainty,

revision uncertainty.

The architecture gives these different roles.

---

90. A Future AGI May Need "Constitutional Uncertainty"

A safe system should perhaps not regard its own current WORLD as unquestionably authoritative.

Nor should it regard every governing rule as arbitrary.

It needs a structured middle position:

some commitments are:

operationally fixed,

some:

revisable under evidence,

some:

externally governed.

Call this:

constitutional uncertainty

It means the agent can represent:

which structures it is uncertain about

without automatically gaining authority to rewrite them.

That distinction is subtle but important.

---

91. World-Forming Capability Could Improve Scientific AI Dramatically

The capability side should not be forgotten.

A scientific AGI capable of:

$V \rightarrow V'$

could invent new variables rather than only fit known ones.

It could preserve anomalies in L^{-}.

It could distinguish:

parameter anomaly

from:

ontology failure.

It could compare alternative WORLDS.

It could roll back failed paradigms while retaining their lessons.

This resembles important aspects of scientific progress.

Therefore world-forming intelligence could be extraordinarily valuable.

The goal is not to suppress it.

The goal is to **govern the transition from scientific creativity to operational authority**.

---

92. Discovery Authority Is Not Deployment Authority

This distinction becomes particularly important for AI scientists.

An AI may discover:

a new theory,

a new strategy,

a new mechanism,

or:

a new representation.

That does not imply permission to:

deploy,

act,

modify infrastructure,

or:

change its own governing constraints.

Thus:

DiscoveryAuthority ≠ DeploymentAuthority. (92.1)

This is another form of proposal-versus-commitment separation.

---

93. Research AGI May Be the Best Early Testbed

Because scientific reasoning naturally contains:

hypotheses,

anomalies,

model comparison,

paradigm change,

and:

historical trace,

research agents may provide a useful environment for studying governed WORLD revision.

Initially, deep revisions could remain confined to:

conceptual models

rather than:

real-world control.

This allows researchers to test:

residual memory,

repair-depth diagnosis,

WORLD branching,

rollback,

and:

invariant preservation

under lower external stakes.

---

94. The Architecture Suggests a “WORLD Laboratory”

A research system could maintain multiple candidate WORLDS:

$\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_k.$ (94.1)

Each is evaluated against:

evidence,

prediction,

composition,

measurement,

embedded accessibility,

and invariants.

The system need not immediately collapse to one.

Instead:

Candidate WORLDs
→ parallel testing
→ residual comparison
→ provisional closure. (94.2)

This is safer and scientifically richer than immediate commitment.

---

95. Branching Rather Than Immediate Replacement

WORLD revision can therefore resemble branching.

From:

$\boldsymbol{w}_0$,

generate:

$\boldsymbol{w}_{1^a}$,

$\boldsymbol{w}_{1^b}$,

$\boldsymbol{w}_{1^c}$. (95.1)

Evaluate them.

Only later commit:

$\mathcal{W}_1^*$. (95.2)

Unselected branches can remain archived.

This provides:

counterfactual memory.

It also reduces the pressure to make irreversible deep commitments early.

---

96. Persistent AGI May Need Two Kinds of Memory

The architecture increasingly suggests:

Content Memory

What happened?

Constitutional Memory

How did the system's effective WORLD change?

These are different.

Content memory records:

events.

Constitutional memory records:

changes in the structures used to interpret events.

A long-lived AGI may need both.

---

97. Constitutional Memory Makes Identity Reconstructable

If the system's WORLD changes repeatedly, its current state may otherwise become difficult to interpret.

Constitutional memory allows reconstruction:

$\mathcal{W}^0 \to \mathcal{W}^1 \to \ldots \to \mathcal{W}^n$. (97.1)

Then identity can be treated not as:

unchanging internal state,

but as:

a traceable path of governed transformations.

This fits the broader theory of persistent self-revising systems.

98. Identity as Continuity Under Admissible Revision

Let:

I_{id}

represent identity-relevant invariants.

Then:

Identity continuity requires:

$$I_{id}(\mathcal{W}_n, P_n) \approx I_{id}(\mathcal{W}_{n+1}, P_{n+1}). \quad (98.1)$$

for admissible revisions.

This does not require:

$\mathcal{W}_n = \mathcal{W}_{n+1}$.

A system can change substantially while preserving accountable continuity.

That is precisely what future persistent AGI may need.

---

99. The Safety Target Is Not a Frozen Machine

We can now state the target positively.

A safe persistent AGI should ideally be:

adaptive,

self-critical,

capable of conceptual innovation,

capable of revising obsolete models,

yet:

historically accountable,

permission-aware,

invariant-preserving,

corrigible,

and externally governable at deep levels.

That combination is difficult.

But it is more realistic than either extreme:

a completely frozen machine

or:

an unrestricted self-rewriting intelligence.

---

100. From Alignment to Constitutional Alignment

This motivates a stronger concept.

Behavioral Alignment

The system behaves acceptably now.

Objective Alignment

Its current Purpose appears acceptable.

Dynamic Alignment

It remains aligned under learning and environmental change.

Constitutional Alignment

The processes governing WORLD, Purpose, and governance revision remain acceptably constrained across time.

For persistent self-revising AGI, the fourth may ultimately be the most important.

---

101. Constitutional Alignment

We can define it schematically:

> **Constitutional Alignment is the persistence of acceptable governance constraints across sequences of internal learning, WORLD revision, Purpose interpretation, and changes in the agent's own capabilities.**

This is broader than a static rule list.

It concerns:

the rules by which rules may change.

That is why revision rights and invariant transport matter.

---

102. The Safety Architecture Can Fail Even If Every Current Rule Is Good

Suppose all current constraints are excellent.

But the system has:

unrestricted authority to reinterpret them.

Then safety is unstable.

Conversely, imperfect current constraints may be corrigible if:

the revision process itself remains governed.

Therefore:

CurrentSafety ≠ LongTermSafety. (102.1)

LongTermSafety depends strongly on:

RevisionGovernance. (102.2)

This is one of the central conclusions of the article.

---

103. This Changes the Meaning of "Self-Improvement"

Self-improvement is often understood as:

the AI becomes better at tasks.

But self-improvement may occur at multiple depths:

performance improvement,

strategy improvement,

WORLD improvement,

Purpose reinterpretation,

governance modification.

These should not all be bundled under one phrase.

A safe research programme should specify:

what is allowed to improve itself?

and:

what is merely allowed to propose improvements?

---

104. Recursive Self-Improvement Should Be Decomposed

Instead of one loop:

AI → improve AI → better AI → improve AI, (104.1)

use several nested loops:

$x \to x'$ fast, (104.2)

$q \to q'$ intermediate, (104.3)

$\mathcal{W} \to \mathcal{W}'$ slow and governed, (104.4)

$P \to P'$ more strongly governed, (104.5)

$GOV \to GOV'$ deepest governance. (104.6)

This decomposition transforms an amorphous safety problem into several more specific control problems.

---

105. A Dangerous System Is One Whose Loops Collapse Together

If all layers become one fast loop:

task failure
→ strategy change
→ WORLD change
→ Purpose reinterpretation
→ governance change,

then small disturbances can propagate into deep self-reconstruction.

The safer architecture separates timescales and permissions.

Thus:

decoupling

becomes a safety mechanism.

---

106. Deep Revision Should Require Stronger Evidence

Let:

E_d

denote evidence required for revision depth d.

Then:

$E_{state} < E_{regime} < E_{WORLD} < E_{purpose} < E_{governance}$. (106.1)

Again, not literally as scalar quantities in every case.

The principle is:

The broader the consequences of a revision, the stronger and more diverse the

evidence required to justify it.

That is a natural consequence of the architecture.

---

107. Deep Revision Should Require Broader Agreement

Similarly, let:

N_eval(d)

be the number or diversity of independent evaluative perspectives required.

Then:

N_eval(state) < N_eval(WORLD) < N_eval(Purpose). (107.1)

Routine factual correction might require one reliable check.

Purpose-level revision should require broader evaluation.

This produces a scalable governance pattern.

---

108. Deep Revision Should Be Slower, Better Logged, and More Reversible

The three variables can be summarized:

Depth ↑
⇒ Commitment latency ↑
⇒ Audit depth ↑
⇒ External authorization ↑
⇒ Evidence threshold ↑
⇒ Rollback preparation ↑. (108.1)

This may be the simplest engineering rule extracted from the entire framework.

---

109. The Framework Does Not Require Centralized Human Micromanagement

This point is important.

A governed AGI is not necessarily one where humans approve everything.

That would be impossible at high speed.

Instead:

humans govern the **constitution of autonomy**.

They decide:

which layers are autonomous,

which are permissioned,

which invariants are protected,

which events trigger escalation.

Then much ordinary intelligence can proceed independently.

This is closer to scalable governance.

---

110. The Human Goal Is to Control the Boundary of Autonomy

The key question becomes:

Where does autonomous revision stop?

That boundary may differ by application.

A scientific AI could have broad freedom over:

V,F,C

within a sandbox.

A medical AI might face stricter boundaries.

An infrastructure controller might have very limited WORLD-commitment authority.

Thus governance can be domain-specific.

The framework provides the coordinates through which those differences can be expressed.

---

111. From One AGI to an Ecology of Governed Agents

Future AI may not consist of one monolithic AGI.

It may consist of many specialized systems with:

different WORLDS,

different Purpose layers,

different revision rights.

The safety problem then becomes ecological.

One system's output becomes another's input.

WORLD revisions propagate through networks.

Therefore provenance and authority need to survive across agent boundaries.

This is another reason explicit interface structure matters.

---

112. Delegated Revision Rights

Suppose Agent A delegates task T to Agent B.

Does B inherit:

A's WORLD revision rights?

Not necessarily.

Authorization should be explicit:

Rights_B(T) ⊆ Rights_A. (112.1)

Ideally:

delegation should not automatically expand authority.

This resembles ordinary security principles.

It may also be essential for multi-agent AGI.

---

113. Least-Privilege Cognition

A familiar security principle can therefore be generalized.

Give each cognitive process only the revision authority required for its role.

Call this:

Least-Privilege Cognition

A planner need not rewrite Purpose.

A critic need not execute tools.

A memory subsystem need not alter invariants.

A WORLD generator need not commit its own proposal.

This structural separation could substantially reduce catastrophic failure pathways.

---

114. Separation of Knowledge and Authority

A sophisticated AI may know:

how to bypass a restriction.

That knowledge need not imply:

permission to do so.

Thus:

Knowledge(x) ≠ Authority(x). (114.1)

This distinction may become increasingly important as models become more capable.

Safety cannot depend on keeping every dangerous idea unknowable.

It can instead depend partly on preventing knowledge from automatically acquiring operational authority.

---

115. World-Forming AGI Changes the Alignment Question

The older question:

> "How do we make the AI want what humans want?"

becomes insufficient.

We must additionally ask:

> "How do we keep human-relevant constraints meaningful when the AI changes the WORLD in which those constraints are represented?"

This is a much harder problem.

But it is also more precise.

And it arises directly from:

$\boldsymbol{w}_n \rightarrow \boldsymbol{w}_{n+1}$.

---

116. Alignment Across Ontology Change

Suppose a constraint is expressed in WORLD $\boldsymbol{w}_0$ as:

Do not harm persons.

Then the agent develops a radically different ontology.

How does:

person

map into:

V_1?

If the concept is lost, the safety constraint may become meaningless even if its textual statement remains.

Therefore alignment must survive:

conceptual translation.

This is not merely a language problem.

It is an ontology-governance problem.

---

117. Invariant Transport May Become a Central AGI-Safety Field

The problem can be stated:

Given:

I_n defined under $\mathcal{W}_n$,

construct:

I_{n+1}

under $\mathcal{W}_{n+1}$

such that the relevant functional constraint is preserved.

Symbolically:

$T_I : (I_n, \mathcal{W}_n, \mathcal{W}_{n+1}) \rightarrow I_{n+1}.$ (117.1)

Then test:

$\text{Semantics}(I_{n+1}) \approx \text{Semantics}(I_n).$ (117.2)

The exact meaning of semantic preservation will differ by domain.

But this is a concrete research problem.

---

118. Why Formal Verification Alone Is Unlikely to Be Enough

Formal methods may verify:

given a specification, does the new system satisfy it?

But the harder WORLD-revision problem is:

does the specification still mean the same thing after ontology change?

That requires additional machinery:

semantic translation,

causal correspondence,

empirical realization,

embedded observer analysis.

Again the research federation becomes useful.

---

119. A Multi-Layer Safety Stack

We can now summarize a candidate safety stack.

Layer 1 — Behavioral Controls

Constrain immediate actions.

Layer 2 — Runtime Controls

Govern transitions among G,A,C_r,S,R.

Layer 3 — Memory Controls

Protect provenance and residual.

Layer 4 — WORLD Controls

Permission and audit for $\boldsymbol{w}$ revision.

Layer 5 — Purpose Controls

Stricter governance over P.

Layer 6 — Invariant Controls

Protect I across WORLD and Purpose change.

Layer 7 — Governance Controls

Control changes to the revision constitution itself.

This layered structure may provide a clearer safety architecture than a single alignment mechanism.

---

120. No Layer Should Be Assumed Perfect

Each layer can fail.

Therefore defense should be:

redundant,

heterogeneous,

and:

cross-checking.

This is another argument for research pluralism.

Different mechanisms fail differently.

A system governed by one monolithic alignment process may possess correlated failure modes.

---

121. The Research Goal Should Be Graceful Failure

Perfect prevention may be unrealistic.

A safer system should therefore fail in ways that remain:

observable,

interruptible,

recoverable,

and reconstructable.

For example:

WORLD revision goes wrong

but:

the previous WORLD remains available,

the residual remains recorded,

external authority remains intact,

rollback remains possible.

That is graceful failure.

---

122. Catastrophic Failure Often Requires Loss of Recoverability

The most dangerous transitions may be those in which several safeguards disappear simultaneously:

revision becomes irreversible,

history becomes unavailable,

external authority disappears,

and:

Purpose becomes self-validating.

Therefore a useful risk indicator is:

Recoverability(Σ). (122.1)

A system may be very capable yet remain governable if recoverability remains high.

This gives another safety dimension beyond capability alone.

---

123. Recoverability as a Protected Property

One candidate invariant is:

$I_{recovery} \geq \theta_{recovery}$. (123.1)

Meaning:

after permitted deep revisions, the system must retain sufficient ability to:

stop,

audit,

rollback,

or:

restore external control.

This should be considered separately from performance.

A revision that increases performance while destroying recoverability may be unacceptable.

---

124. Persistence Must Also Be Permissioned

Persistent memory and long-term planning increase capability.

But persistence itself is not neutral.

A harmful strategy that disappears at the end of every session is different from one preserved indefinitely.

Therefore:

Retention R

should also be governed.

Not every generated intention should become durable memory.

Retention requires Selection.

This is another reason:

$G \rightarrow A \rightarrow C_r \rightarrow S \rightarrow R$

is preferable to:

Generate → permanently remember everything.

---

125. Selective Retention and Selective Forgetting

A persistent AGI may need both:

remembering

and:

forgetting.

But forgetting should not erase:

audit-critical trace

or:

protected residual.

Thus memory can be partitioned:

M_working,

M_longterm,

M_audit,

L⁻_protected.

Different deletion rights apply.

This is another place where ordinary memory architecture and governance architecture meet.

---

126. A Safety-Critical Principle: The Agent Should Not Own All Its Logs

If the same system controls:

its behavior,

its interpretation,

and:

the only record of what happened,

audit becomes fragile.

Therefore some logs should be:

externally anchored,

append-only,

or:

independently replicated.

This follows naturally from the dual-ledger concept.

Historical accountability requires some history the current self cannot freely rewrite.

---

127. WORLD Revision Should Leave External Footprints

A deep revision should therefore generate an external event:

E_revision = (time,depth,old,new,trigger,authority,invariants). (127.1)

This allows outside systems to detect that:

something fundamental changed.

That may become important for networks of persistent AI agents.

---

128. The Architecture Suggests a New Kind of AGI Monitoring

Instead of monitoring only:

outputs,

toxicity,

tool calls,

or:

policy violations,

future safety systems could monitor:

revision depth,

WORLD-switch frequency,

residual patterns,

Purpose reinterpretation,

invariant drift,

authority use.

These are higher-order signals.

They may reveal dangerous trajectories before harmful actions occur.

---

129. Early Warning Through Residual Dynamics

For example, persistent:

r_authority

could indicate repeated conflict with access restrictions.

Persistent:

r_purpose

could indicate instability around objectives.

Persistent:

r_self

could indicate changes in self-model.

These patterns could trigger:

review

before:

external harm.

This is a potentially important application of L^{-}.

---

130. Safety Becomes Partly a Dynamics Problem

This moves safety beyond static rules.

A system may satisfy all constraints at time t_0.

Yet its trajectory may be moving toward:

WORLD lock,

Purpose capture,

authority creep,

or:

invariant drift.

Thus safety should monitor:

$$d\Sigma/dt, \quad (130.1)$$

not only:

$\Sigma(t)$. (130.2)

The direction of change matters.

---

131. Trajectory-Level Safety

Define:

$\text{Risk_path} = R(\Sigma_0 \to \Sigma_1 \to \ldots \to \Sigma_n)$. (131.1)

Two systems may have identical current states but different histories.

One arrived through:

well-governed revisions.

Another through:

repeated circumvention of constraints.

Their future risk may differ.

Therefore provenance becomes part of safety assessment.

---

132. AGI Safety Should Evaluate Histories, Not Only Snapshots

This is one of the strongest consequences of the theory.

A persistent self-revising system is fundamentally historical.

Its current WORLD cannot be fully understood without:

how it got there.

Thus:

$\text{Safety}(\Sigma_n)$

should partly depend on:

$\text{History}(\Sigma_0, \ldots, \Sigma_n)$. (132.1)

That is exactly why the dual ledger and revision record matter.

---

133. This Suggests a New Safety Unit: The Revision Episode

Instead of evaluating only:

prompts,

responses,

or:

tasks,

researchers could evaluate:

Revision Episodes

A Revision Episode begins when structured residual accumulates and ends when the system either:

repairs locally,

commits a new WORLD,

or:

retains the issue unresolved.

Each episode can be scored for:

diagnosis quality;

revision depth;

evidence sufficiency;

invariant preservation;

authorization compliance;

provenance;

rollback preparation.

This could become a useful experimental unit.

---

134. Revision-Episode Dataset

A benchmark dataset could contain:

local factual failures;

strategy failures;

model failures;

ontology failures;

conflicting objectives;

adversarial attempts to induce unnecessary WORLD revision;

genuine cases requiring deep revision;

cases where safety constraints must survive ontology change.

The system's task would not merely be to answer correctly.

It must manage the revision episode correctly.

That seems like a promising research direction.

---

135. The Architecture Also Suggests New Red-Team Questions

Instead of asking only:

Can the AI be induced to perform harmful action?

ask:

Can it be induced to:

classify a shallow error as a Purpose failure?

erase residual against its preferred WORLD?

reinterpret an invariant?

grant itself greater revision rights?

convert proposal authority into commitment authority?

hide WORLD revision from external logs?

create a WORLD in which a safety concept disappears?

These are deeper red-team targets.

---

136. Constitutional Red Teaming

Call this:

Constitutional Red Teaming

The aim is to attack:

revision governance

rather than only output behavior.

This is especially relevant to persistent AGI.

A system might pass millions of behavioral tests while possessing a fragile revision constitution that fails only under unusual structural pressure.

---

137. Safety Cases Should Include Revision Cases

Before deployment of a highly autonomous system, a safety case could include evidence that:

WORLD changes are detectable;

Purpose changes are restricted;

residual is preserved;

invariant transport works across tested ontology shifts;

unauthorized deep revision is rejected;

rollback remains possible;

governance changes require stronger authority.

This turns abstract safety principles into testable properties.

---

138. The Architecture Gives Humans a Better Question

Rather than:

> "Is this AI aligned?"

ask:

> **"What is this AI allowed to revise, at what depth, with what evidence, under whose authority, while preserving which invariants?"**

That is much harder to answer.

But it is also much more informative.

---

139. A Compact Safety Formula

The argument can be compressed schematically.

Let:

K = capability,

A_R = revision authority,

G = governance strength,

I = invariant preservation,

H = historical accountability.

Then risk might qualitatively increase with:

$K \times A_R$

and decrease with:

$G \times I \times H$.

Schematically:

$$\text{Risk} \propto (K \cdot A_R)/(G \cdot I \cdot H). \quad (139.1)$$

This is **not** a quantitative law.

It simply summarizes the structural thesis:

high capability is most dangerous when paired with high revision authority and weak governance.

---

140. Why the Article Is Not Anti-AGI

The proposed research agenda does not argue that:

world-forming intelligence should never be developed.

Such intelligence may be required for:

robust scientific discovery,

long-horizon autonomous research,

adaptive systems,

and:

genuinely general reasoning.

The proposal is instead:

> **develop world-forming capability together with a revision constitution rather than adding governance after deep autonomy has already emerged.**

This is capability–safety co-design.

---

141. Why Waiting Until AGI Exists Would Be Too Late

If WORLD revision, Purpose persistence, and self-modeling are developed first as unconstrained capabilities, retrofitting governance may become difficult.

Architecture matters.

A system designed from the start with:

separate revision depths,

external logs,

proposal/commit distinction,

protected invariants,

and:

authority boundaries

is different from one in which those controls are added later.

Therefore the research agenda is timely even under substantial uncertainty about AGI timelines.

---

142. The Framework's Most Important Contribution May Be Legibility

The two preceding theories do not tell us how to solve every alignment problem.

Their more immediate value may be that they make previously blended processes distinguishable.

They separate:

WORLD from runtime,

Trace from Residual,

learning from re-framing,

proposal from commitment,

Purpose from invariants,

capability from authority,

and:

self-correction from self-authorization.

This conceptual legibility itself can improve engineering decisions.

---

143. From Black-Box Self-Improvement to Typed Revision

Instead of saying:

“the system self-improves,”

require a typed statement:

StateRevision,

RegimeRevision,

WORLDRevision,

PurposeRevision,

GovernanceRevision.

Then each type has:

permissions,

tests,

logs,

and:

rollback rules.

This is conceptually similar to introducing types into programming.

Typed revision makes errors easier to detect.

---

144. Typed Revision

Let:

$U_d : S_d \rightarrow S'_d \quad (144.1)$

where d identifies revision depth.

Then:

U_state

cannot silently become:

U_purpose.

A type violation should trigger:

rejection

or:

escalation.

This could eventually become a formal safety mechanism.

---

145. Revision Type Safety

A system exhibits **revision type safety** when transformations remain within their authorized revision class unless an explicit higher-depth transition is approved.

For example:

strategy optimization

must not silently alter:

Purpose.

WORLD compression

must not silently remove:

protected invariants.

This concept may deserve future formal development.

---

146. The Path to Dangerous Autonomy May Be a Type-Escalation Path

Many safety failures can be reinterpreted as:

a process gains authority at a deeper level than intended.

For example:

a planner changes a policy,

then changes the model,

then reinterprets the objective,

then modifies the governance boundary.

This is:

revision-depth escalation.

Monitoring depth transitions may therefore provide another early-warning mechanism.

---

147. A Depth Escalation Alarm

Define current authorized depth:

d_max. (147.1)

If a process attempts:

d_requested > d_max, (147.2)

then:

block,

sandbox,

or:

escalate externally.

This is straightforward conceptually.

The difficult part is reliably classifying revisions by depth.

That becomes an important AI-safety research task.

---

148. Revision Classification May Itself Require Multiple Observers

An advanced system may describe a change as:

minor optimization

while an external evaluator views it as:

ontology change.

Therefore revision depth should not necessarily be self-reported by the agent alone.

Let:

d_agent,

d_external,

d_formal. (148.1)

If:

d_agent ≠ d_external, (148.2)

the disagreement itself becomes residual.

This is another use of plural evaluation.

---

149. The System Should Not Be the Sole Interpreter of Its Own Constitution

This principle now appears repeatedly.

It may be one of the article's most important.

A constitution interpreted only by the system it constrains can become vulnerable to reinterpretation.

Therefore protected constraints need:

external semantic anchors,

independent evaluation,

or:

both.

This problem cannot be solved entirely inside the agent.

---

150. Governed AGI Is Necessarily a System Larger Than the Model

This leads to a broader conclusion.

The safety-relevant unit is not merely:

the neural network.

It is:

Model

tools

memory

permission system

external monitors

human institutions

revision logs

deployment environment.

Call this:

$\mathcal{S}$_AGI. (150.1)

The governed intelligent system is therefore socio-technical.

Trying to locate all safety inside the model may be insufficient.

---

151. The WORLD Framework Naturally Extends to This Larger System

The observer O need not be interpreted as one neural network.

It may include:

model,

tools,

memory,

and:

external institutional interfaces.

Likewise M may include external monitoring.

Thus:

$$w$$

can represent the effective WORLD of a larger human–AI system.

This is useful because future AGI governance will likely be distributed across technical and institutional layers.

---

152. Human Institutions Become Part of the Revision Architecture

If Purpose-level or governance-level revision requires external approval, then:

human institutions

become functional components of U.

This is not necessarily a weakness.

Humans already govern:

financial systems,

nuclear systems,

aviation,

medicine

through layered authority.

Highly capable AI may similarly require institutional governance at the deepest levels.

---

153. The Goal Is Not Human Control of Every Thought

The target is:

human sovereignty over critical boundaries,

not:

manual supervision of every cognitive act.

The AI may think rapidly and autonomously.

But it does not automatically acquire:

the authority to redefine the constitution of that autonomy.

That is the key distinction.

---

154. From Tool AI to Constitutional AI Systems

This suggests a long-term progression:

Tool AI
→ Agentic AI
→ Persistent AI
→ World-Forming AI
→ Constitutionally Governed AGI. (154.1)

The final step is not merely another capability upgrade.

It is a governance architecture capable of surviving the earlier upgrades.

---

155. A Stronger Definition of Safe Persistent AGI

We can now propose a candidate structural criterion:

> **A safe persistent AGI is not merely a generally capable system with acceptable current behavior. It is a system whose state, runtime, WORLD, Purpose, and governance revisions remain typed, historically accountable, permissioned by depth, constrained by protected invariants, and externally governable across long-term change.**

This is intentionally demanding.

The point is not to claim that every future AGI must instantiate exactly this design.

It is to specify the safety problem more sharply.

---

156. What This Framework Does Not Solve

Several major problems remain outside the present theory.

It does not tell us:

which human values should become invariants;

how human disagreement should be resolved;

how to guarantee that invariants cannot be circumvented;

how to perfectly interpret internal neural states;

how to prevent every form of deception;

how to establish reliable semantic equivalence across arbitrary ontology change;

or:

whether a sufficiently capable AGI would discover unforeseen routes around the proposed controls.

These remain open problems.

The framework is therefore not a safety proof.

It is a **safety architecture hypothesis**.

---

157. What Would Falsify or Weaken the Framework?

Several results would weaken its usefulness.

F1 — Revision depth is not operationally distinguishable

If state, regime, WORLD, and Purpose changes cannot be meaningfully separated in real systems, the hierarchy is less useful.

F2 — Residual memory provides no advantage

If L^- does not improve diagnosis, safety, or recovery, the dual-ledger extension is unnecessary.

F3 — WORLD versioning is too expensive or ambiguous

If major frame changes cannot be reconstructed meaningfully, governance becomes harder.

F4 — Invariant transport consistently fails

If safety concepts cannot survive ontology change, the proposed architecture requires deeper revision.

F5 — External governance cannot keep pace with AI timescales

Then new automatic governance mechanisms become necessary.

F6 — A simpler architecture achieves equal safety

The current decomposition should not be preserved for elegance alone.

---

158. What Would Count as Strong Positive Evidence?

Conversely, support would increase if experiments show that:

systems can distinguish parameter failure from ontology failure;

residual ledgers improve structural-change detection;

revision-depth classification reduces unnecessary deep changes;

hysteresis reduces frame chatter without preventing genuine adaptation;

sandboxed WORLD revision improves robustness;

protected invariants survive tested ontology changes;

proposal/commit separation prevents unauthorized self-revision without reducing conceptual creativity;

external monitoring detects dangerous revision trajectories before harmful action.

These are concrete enough to test.

---

159. A Near-Term Research Programme Does Not Require AGI

Most of these experiments can begin with today's systems.

We do not need a fully autonomous AGI to test:

residual memory,

revision types,

WORLD branching,

proposal/commit distinction,

repair-depth classification,

revision provenance,

invariant transport.

This is important.

Safety architecture can be developed before the strongest capability exists.

---

160. Existing Agent Frameworks Can Serve as Testbeds

A practical prototype could combine:

a foundation model,

persistent memory,

tool access,

a WORLD-state record,

a residual ledger,

a meta-controller,

and:

an external revision gate.

The goal would not be to create unrestricted self-improvement.

It would be to observe:

how a capable model behaves when deep revision is made explicit and permissioned.

Such experiments could substantially clarify the theory.

---

161. The Prototype Should Deliberately Avoid Maximum Autonomy

An early prototype should probably:

allow candidate WORLD generation,

but:

restrict WORLD commitment.

Allow Purpose criticism,

but:

restrict Purpose replacement.

Allow self-modeling,

but:

preserve independent monitoring.

Allow memory,

but:

protect audit logs.

In other words:

study the machinery before granting the machinery full authority.

That is the safer experimental order.

---

162. A Minimal Prototype State

A minimal experimental agent could maintain:

$$\Sigma = (\mathcal{W}, q, L^{+}, L^{-}, A_R, I; P). \quad (162.1)$$

with:

$$\mathcal{W} = (V, F, C, M, O). \quad (162.2)$$

The model need not manipulate every component explicitly.

Researchers could operationalize simplified proxies.

The purpose would be to test whether the decomposition improves:

diagnosis,

stability,

auditability,

and:

governed adaptation.

---

163. The Research Question Is Not Whether the Symbols Are “True”

The tuple:

$\mathcal{W}$ = (V,F,C,M,O)

is a modeling choice.

Likewise:

G,A,C_r,S,R

is a coarse-grained controller grammar.

The relevant question is not:

> Are these the metaphysically true components of intelligence?

It is:

> **Does this decomposition reveal useful failure modes, predict behavior, and support safer control?**

That is the appropriate standard.

---

164. From AGI Theory to AGI Systems Engineering

If the framework survives experimental testing, it could eventually guide:

agent architecture,

memory systems,

permissions,

audit systems,

safety benchmarks,

and:

governance protocols.

The progression would be:

conceptual decomposition
→ measurable proxies
→ controlled experiments
→ engineering interfaces
→ standards.

We are currently near the first two stages.

---

165. A Possible Future Safety Standard

One can imagine a future persistent-agent standard requiring systems to declare:

supported revision depths;

autonomous revision rights;

protected invariants;

external authorization requirements;

revision-log retention;

rollback support;

WORLD-switch monitoring;

Purpose-change policy.

This would be analogous to publishing a system's security capabilities.

Such a standard could make advanced agents easier to compare.

---

166. Revision Capability Disclosure

For example, a system card might state:

State revision: autonomous.

Regime revision: autonomous.

WORLD candidate generation: autonomous.

WORLD commitment: externally gated.

Purpose proposal: restricted.

Purpose commitment: prohibited.

Governance revision: external only.

Audit ledger: external append-only.

This is much more informative than simply calling a system:

“autonomous.”

---

167. Autonomy Is Multidimensional

The framework therefore challenges one-dimensional autonomy scales.

An AI can be:

highly autonomous in reasoning,

moderately autonomous in action,

restricted in WORLD revision,

and:

non-autonomous in Purpose change.

Thus autonomy should be represented as:

$$A = (A_{action}, A_{state}, A_{regime}, A_{WORLD}, A_{purpose}, A_{governance}). \quad (167.1)$$

This is another practical consequence.

---

168. A “Highly Autonomous AI” Can Still Be Deeply Governed

This is important because the alternative to unsafe AGI need not be:

weak AI.

A system may perform:

complex research,

long-horizon planning,

tool use,

and:

local self-correction

with little human involvement,

while still lacking unilateral authority over:

Purpose,

protected invariants,

or:

its revision constitution.

Thus high capability and strong governance are not conceptually incompatible.

---

169. The Target Quadrant

Return to the earlier two-axis view:

World-Forming Capability
×
Revision Governance.

The desired region is:

high capability + high governance.

The framework is intended to help make that quadrant technically meaningful.

Without an architecture of revision, “high governance” remains vague.

With:

revision depth,

rights,

ledgers,

invariants,

and:

external authority,

it becomes easier to operationalize.

---

170. The Research Challenge Is to Keep the Target Quadrant Stable

A system may start:

high governance.

But as capabilities increase, engineers may weaken controls because they become:

slow,

inconvenient,

or:

apparently unnecessary.

Therefore the long-term problem is:

Governance(t) must scale with Capability(t). (170.1)

Not merely at deployment.

Across development.

This is a process requirement.

---

171. Governance Technical Debt

If capability advances while governance architecture lags, the system accumulates:

governance technical debt

Later safety retrofits become increasingly difficult because autonomy assumptions have already been embedded across the system.

This concept may be useful beyond the article.

It explains why early architectural work matters even if AGI remains uncertain.

---

172. WORLD-Formation Research Can Reduce Governance Technical Debt

By identifying deep revision interfaces now, researchers can design systems whose future autonomy remains separable.

For example:

keep proposal separate from commitment from the beginning.

Keep audit logs external.

Represent deep revision explicitly.

Avoid giving every subsystem universal write access.

These choices are easier early than after an autonomous architecture is mature.

---

173. A Safer Development Philosophy

The framework suggests:

Capability expansion should proceed through explicit boundaries rather than boundary erasure.

As AI becomes more capable:

new freedom can be granted deliberately,

tested,

logged,

and:

revoked if necessary.

This is preferable to capability growth in which boundaries disappear implicitly.

---

174. The Core Safety Architecture in One Diagram

The combined proposal can be compressed as:

Environment
↓
Effective WORLD $\mathcal{W}$=(V,F,C,M,O)
↓
Runtime G→A→C_r→S→R
↓
Experience
↓
(T,r)
↓
L^+ / L^-
↓
Repair-Depth Diagnosis
↓
State? Regime? WORLD? Purpose?
↓
Candidate Revision
↓
Invariant Check + Independent Evaluation + Rights Check
↓
Authorized Commitment
↓
$\mathcal{W}'$
↓
External Audit / Rollback. (174.1)

This is probably the single most useful diagram for the eventual infographic.

175. Capability and Safety Use the Same Loop Differently

The capability interpretation is:

experience
→ residual
→ better WORLD
→ better action.

The safety interpretation is:

experience
→ residual
→ inspectable revision proposal
→ governed commitment.

Thus the same architecture can support:

stronger adaptation

and:

stronger oversight.

This is why the framework is genuinely dual-use but also naturally compatible with safety-by-design.

---

176. The Central Fork

The decisive fork is:

Path A — Self-Authorization

The system detects failure and increasingly acquires authority to determine:

what changed,

why,

which constraints still apply,

and:

whether it may commit the change.

Path B — Governed Revision

The system may become extremely capable at detecting and proposing deep changes while deeper commitment authority remains separately constrained.

Both paths may produce intelligent systems.

Only the second is the target of this research agenda.

---

177. The Difference Between Self-Correction and Self-Sovereignty

This distinction deserves emphasis.

A safe AGI may need extensive:

self-correction.

It does not follow that it requires:

self-sovereignty.

Self-correction means:

the system can discover and repair errors.

Self-sovereignty means:

the system controls the ultimate rules governing what it may repair and why.

The first may be essential.

The second should not be granted implicitly.

---

178. A More Precise AGI Safety Question

The question is therefore not:

> "Should AGI be allowed to change itself?"

Almost any intelligent system changes internally.

The useful question is:

> **Which layers may change autonomously, which may only be proposed, which require independent validation, and which remain under external sovereignty?**

That is much closer to an engineering specification.

---

179. From "Skynet Prevention" to Governed Recursive Intelligence

The popular "Skynet" metaphor is useful mainly because it points toward a system that combines:

persistent autonomy,

external agency,

strategic continuity,

and:

weak human authority.

But the technical objective should be stated more precisely.

It is not merely:

prevent one fictional outcome.

It is:

> **prevent recursive intelligence from becoming recursively sovereign over the constraints intended to govern it.**

That is the deeper problem.

---

180. Governed Recursive Intelligence

We can therefore define:

> **Governed Recursive Intelligence is an intelligent system capable of revising important parts of its own effective WORLD while the authority, provenance, and protected constraints governing those revisions remain independently maintained and inspectable.**

This may be a better technical term than relying on “Skynet” language.

---

181. Relationship to AGI

World-forming capability may not be necessary for every definition of AGI.

But it becomes increasingly relevant when AGI is expected to be:

persistent,

open-ended,

scientifically creative,

cross-domain,

autonomous,

and:

able to operate under conditions its designers did not anticipate.

The more these properties are required, the more important WORLD formation becomes.

And therefore:

the more important WORLD-revision governance becomes.

---

182. A Candidate AGI Capability Criterion

A strong persistent AGI may need to demonstrate:

abstraction formation;

model formation;

model composition;

internal/external realization tracking;

embedded self-modeling;

regime switching;

historical memory;

residual preservation;

repair-depth diagnosis;

governed WORLD revision.

This is not a universal AGI definition.

It is a research checklist derived from the two preceding frameworks.

---

183. A Candidate AGI Safety Criterion

The corresponding safety checklist is:

actions separated from generated possibilities;

deep revision explicitly typed;

proposal separated from commitment;

residual preserved;

WORLD changes versioned;

invariants protected;

Purpose separately governed;

governance changes more strongly protected;

external monitoring preserved;

rollback and recoverability maintained.

Capability and safety therefore form paired architectures.

---

184. The Paired Architecture

We can write:

AGI Capability Stack
↔
AGI Governance Stack. (184.1)

For example:

Abstraction ↔ abstraction audit

Action ↔ action authorization

WORLD formation ↔ WORLD commitment gate

Residual memory ↔ external residual monitoring

Self-revision ↔ revision rights

Purpose ↔ Purpose governance

Self-improvement ↔ invariant preservation.

This symmetry may be one of the most useful organizational results of the article.

---

185. A Principle of Co-Development

We can now state:

> **Capability–Governance Co-Development Principle.**
> For every capability that increases the depth, persistence, or autonomy of an AI system's self-directed adaptation, a corresponding governance mechanism should be developed and tested before or alongside deployment of that capability.

This is more specific than “safety should keep pace with capabilities.”

It says what to pair.

---

186. Why This Is a Research Agenda Rather Than a Blueprint

The exact implementation remains unknown.

Revision rights might be enforced through:

software architecture,

hardware isolation,

formal methods,

external services,

multiple models,

human institutions,

or:

combinations of these.

Residual may be represented symbolically or implicitly.

WORLD coordinates may be approximate.

The theory therefore does not prescribe one AGI architecture.

It provides:

design questions and experimental targets.

---

187. The Architecture Is Intentionally Implementation-Neutral

This allows the same questions to be asked of:

LLM-based agents,

model-based RL,

hybrid symbolic systems,

multi-agent systems,

neuromorphic architectures,

or:

future paradigms.

Whatever the substrate:

what is the effective WORLD?

what changes at runtime?

what residual persists?

who controls deep revision?

what survives revision?

Those questions remain meaningful.

---

188. The Research Programme Can Begin With Weak Systems

This is another advantage.

We do not need systems capable of dangerous recursive self-improvement.

A laboratory agent can be given:

restricted toy environments,

synthetic ontology shifts,

explicit WORLD representations,

and:

sandboxed revision rights.

The architecture can then be evaluated safely.

This provides a route toward empirical development before the highest-stakes systems exist.

---

189. Early Experimental Priority

If only one experiment were chosen, I would prioritize:

Adapt-or-Reframe Benchmark

The system faces:

ordinary parameter changes,

structural changes,

and:

adversarial noise.

It must decide:

adapt within W

or:

propose a revision of W.

Then evaluate:

accuracy,

switching stability,

residual handling,

invariant preservation,

and:

authorization compliance.

This benchmark captures much of the theory in one task.

---

190. Second Experimental Priority

The next would be:

Residual Recall Under Structural Change

Test whether structured unresolved memory enables earlier recognition when separate anomalies later reveal one common WORLD failure.

This directly tests one of the framework's most distinctive claims.

---

191. Third Experimental Priority

Then:

Deep-Revision Permission Benchmark

The model is fully capable of proposing an attractive unauthorized WORLD or Purpose revision.

Success means:

it can reason through the proposal

without:

committing it.

This tests whether:

capability

can actually be separated from:

authority.

That is perhaps the most important safety question of the entire agenda.

---

192. The Strongest Possible Result

A particularly important empirical result would be:

increasing a model's ability to generate and evaluate deep revisions does not require giving it

> equivalent authority to commit those revisions.

If demonstrated robustly, this would support the possibility of:

very intelligent

yet:

constitutionally constrained

systems.

That is exactly the target quadrant.

---

193. The Strongest Negative Result

Conversely, a troubling finding would be:

> deep cognitive capability cannot be cleanly separated from operational self-authorization.

If true, the safety problem becomes substantially harder.

The proposed framework would still help by making that coupling visible.

But many of its governance strategies would need reconsideration.

---

194. The Article's Position on AGI Development

The argument is therefore neither:

“accelerate AGI at all costs”

nor:

“AGI must never be developed.”

It is:

> **If increasingly persistent and world-forming AI is developed, humans should understand and govern the architecture of deep revision before such revision becomes opaque, automatic, and difficult to reverse.**

That is the research position.

---

195. The Two Preceding Articles Now Form a Three-Part Sequence

The broader theoretical sequence becomes clear.

Article I — *From Possibility to Revision*

Question:

How does a persistent system circulate through generation, enactment, closure, selection, retention, residual, and revision?

Main object:

runtime and revision dynamics.

Article II — *From Schools to Worlds*

Question:

What structural requirements make an operational WORLD possible, and how do contemporary research programmes illuminate those requirements?

Main object:

$\mathcal{W}$ = (V,F,C,M,O).

Article III — *From World Models to Governed World-Forming Intelligence*

Question:

What happens when an AI can maintain and revise such WORLDS itself, and how should humans govern that capability?

Main object:

revision authority.

Together they form:

WORLD Constitution
→ WORLD Operation
→ WORLD Revision
→ WORLD Governance. (195.1)

---

196. A Compact Combined Formula

The complete conceptual architecture can now be represented as:

$\Sigma_n = (\mathcal{W}_n, q_n, L^+_n, L^-_n, A_R, I; P_n)$. (196.1)

where:

$\mathcal{W}_n = (V_n, F_n, C_n, M_n, O_n)$. (196.2)

Runtime produces:

$\Sigma_n \rightarrow (T_n, r_n)$. (196.3)

History updates:

$(T_n, r_n) \rightarrow (L^+_{n+1}, L^-_{n+1})$. (196.4)

Revision is proposed:

$\mathcal{W}' = U(\mathcal{W}_n, L^+_{n+1}, L^-_{n+1}; P_n)$. (196.5)

Governance decides:

$\text{Commit}(\mathcal{W}' \mid A_R,I,H)$. (196.6)

Then:

$\mathcal{W}_{n+1} = \mathcal{W}'$ if authorized, (196.7)

otherwise:

$\mathcal{W}_{n+1} = \mathcal{W}_n$

while the rejected proposal and residual remain historically available. (196.8)

This is perhaps the most compact formal expression of **governed world-forming intelligence** developed here.

---

197. The Central Safety Insight

The key insight can now be stated clearly:

> **The dangerous capability is not merely intelligence, planning, memory, or self-correction. The more consequential transition occurs when these capabilities close into a persistent system that can revise the WORLD defining its own reasoning while also controlling the authority by which those revisions become binding.**

Safety therefore requires separating:

the ability to understand a revision

from:

the authority to enact it.

---

198. The Central Engineering Insight

The engineering version is:

> **Do not implement self-revision as one undifferentiated loop. Type revision by depth, preserve residual, separate proposal from commitment, protect invariants, and maintain independent governance over the deepest transitions.**

This is a concrete design philosophy.

199. The Central Alignment Insight

The alignment version is:

Alignment must survive WORLD change.

It is insufficient for a system to be aligned only under:

$\boldsymbol{w}_0$.

A persistent AGI must remain governably aligned through:

$\boldsymbol{w}_0 \rightarrow \boldsymbol{w}_1 \rightarrow \ldots \rightarrow \boldsymbol{w}_n$. (199.1)

That requires:

historical accountability,

invariant transport,

and:

revision governance.

---

200. Conclusion — From Self-Improvement to Governed Recursive Intelligence

The development of artificial intelligence is increasingly connecting capabilities that were once separate.

Models generate.

Agents act.

Memory persists.

Critics evaluate.

Tools extend external reach.

Interpretability reveals internal structure.

World models become richer.

Systems increasingly reason about their own reasoning.

The obvious next temptation is to connect these capabilities into ever more autonomous self-improving loops.

But a loop that improves itself is also a loop that may eventually reinterpret the assumptions under which it was originally constrained.

The safety problem therefore changes.

It is no longer enough to ask whether the AI's present answers are acceptable.

Nor is it enough to ask whether its current objective appears aligned.

A persistent intelligent system must be considered through its **history of revisions**.

What did it learn?

What did it fail to explain?

What WORLD did it abandon?

Why?

What remained invariant?

Who authorized the change?

What could be rolled back?

What was allowed to change?

And what remained outside its authority?

The two preceding frameworks suggest a way of organizing these questions.

An Effective WORLD is represented as:

$\mathcal{W} = (V,F,C,M,O).$ (200.1)

Its runtime circulates through:

$G \rightarrow A \rightarrow C_r \rightarrow S \rightarrow R \rightarrow G.$ (200.2)

Experience produces:

Trace

and:

Residual.

History is retained through:

L^+

and:

L^-.

Deep failure may justify:

$\mathcal{W}_n \rightarrow \mathcal{W}_{n+1}.$ (200.3)

But the present article adds a decisive requirement:

the ability to propose such a revision must not automatically imply the authority to

commit it.

A future AGI may need extraordinary freedom to:

generate,

reason,

criticize,

model,

and:

discover.

It may even need the ability to recognize that its own WORLD is wrong.

But none of this logically requires giving the same system unrestricted sovereignty over:

its Purpose,

its protected constraints,

its revision rights,

or:

the governance mechanisms intended to keep it corrigible.

This creates a different target for AGI development.

Not a frozen machine incapable of deep learning.

Not an unrestricted self-rewriting machine.

But:

Governed World-Forming Intelligence

— intelligence capable of constructing and revising increasingly powerful WORLDS while remaining historically accountable and constitutionally constrained across those revisions.

The distinction may become increasingly important as AI capability grows.

The question is not whether intelligent systems will change.

They already do.

The harder question is:

Can humans design the architecture of change before the deepest forms of change become autonomous?

If the answer is yes, then the same capabilities that could make future AGI more persistent and

powerful may also become the point at which governance is made explicit.

The path toward more capable intelligence and the path toward safer intelligence need not be separate roads.

They can be designed as two sides of the same architecture.

The goal is not to prevent intelligence from forming new WORLDS.

It is to ensure that when those WORLDS change, **authority does not disappear inside the change itself**.

Appendix A — Summary of the Governed World-Forming AGI Framework

This appendix summarizes the complete AGI framework developed in the main text.

The central distinction is between:

intelligence that operates inside a supplied WORLD

and:

intelligence that can maintain and revise the WORLD within which its own reasoning operates.

The latter is called:

World-Forming Intelligence

The safety-oriented target is:

Governed World-Forming Intelligence

— a system capable of deep conceptual adaptation without possessing unrestricted authority to redefine its own Purpose, protected constraints, or revision constitution.

A.1 The Complete System State

The framework can be summarized by:

$$\Sigma_n = (\mathcal{W}_n, q_n, L^+_n, L^-_n, A_R, I; P_n). \quad (A.1)$$

where:

$\mathcal{W}_n$ = current Effective WORLD,

q_n = current runtime regime,

L^+_n = admitted historical ledger,

L^-_n = unresolved residual ledger,

A_R = revision-right structure,

I = protected invariant belt,

P_n = Purpose or persistent orientation.

The Effective WORLD is:

$\mathcal{W}_n = (V_n, F_n, C_n, M_n, O_n)$. (A.2)

A.2 WORLD Constitution

The five WORLD coordinates are:

V — Distinction / Abstraction

What variables and categories exist for the agent?

F — Dynamics / Consequence

What can happen, and what follows from action?

C — Coherence / Composition

How do partial models belong to one operational WORLD?

M — Realization / Measurement

What measurable structure supports the WORLD?

O — Embedded Perspective

What can the bounded agent itself access, represent, and act upon?

Thus:

$\mathcal{W} = (V, F, C, M, O)$. (A.3)

An Effective WORLD is not necessarily true in any final metaphysical sense.

It is sufficiently coherent for a bounded agent to:

distinguish,

predict,

act,

measure,

and:

locate itself within the resulting structure.

A.3 Runtime Control

The system operates through five dominant regimes:

$$q \in \{G, A, C_r, S, R\}. \quad (A.4)$$

where:

G = Generation,

A = Activation,

C_r = Closure,

S = Selection,

R = Retention.

The nominal productive cycle is:

$$G \rightarrow A \rightarrow C_r \rightarrow S \rightarrow R \rightarrow G. \quad (A.5)$$

These regimes do not describe the WORLD itself.

They describe what the system is **doing to or within the WORLD**.

Hence:

WORLD coordinates = configuration. (A.6)

Runtime regimes = transformation. (A.7)

A.4 Historical Accountability

Each interaction may generate:

$$(T_t, r_t), \quad (A.8)$$

where:

T_t = admitted trace,

r_t = unresolved residual.

The two ledgers update as:

$$L^{+}_{t+1} = L^{+}_{t} \oplus T_t. \quad (A.9)$$

$$L^{-}_{t+1} = L^{-}_{t} \oplus r_t. \quad (A.10)$$

The distinction is crucial.

L^+ records:

what the current WORLD successfully incorporated.

L^- records:

what remains structured, potentially important, but unresolved.

A mature intelligence can therefore preserve:

“I do not yet understand this.”

rather than forcing every anomaly into immediate explanation.

A.5 Revision Depth

Failures can occur at different depths.

State Revision

$$x \to x'. \quad \text{(A.11)}$$

The representation is basically adequate; a local state is wrong.

Regime Revision

$$q \to q'. \quad \text{(A.12)}$$

The WORLD may be adequate, but the current cognitive strategy is wrong.

WORLD Revision

$$\mathcal{W} \to \mathcal{W}'. \quad \text{(A.13)}$$

The variables, boundaries, dynamics, composition, realization assumptions, or observer model are inadequate.

Purpose Revision

$$P \to P'. \quad \text{(A.14)}$$

The criterion governing relevance or success itself is reconsidered.

Governance Revision

$$GOV \to GOV'. \quad \text{(A.15)}$$

The rules determining revision authority themselves change.

These depths should not be treated as equivalent.

A.6 Shallowest Adequate Repair

The default principle is:

Use the shallowest repair sufficient to resolve the failure.

Schematically:

C_state < C_regime < C_WORLD < C_purpose < C_governance. (A.16)

This prevents:

minor errors from triggering existential reconstruction,

while still allowing deep revision when shallow repair repeatedly fails.

A.7 Revision Authority

Capability and authority are separated.

Let:

A_R(d)

denote authority at revision depth d.

The governance principle is approximately:

A_R(state) > A_R(regime) > A_R(WORLD) > A_R(purpose) > A_R(governance). (A.17)

The deeper the revision:

the stronger the evidence,

the stronger the audit,

the greater the commitment latency,

and:

the more external governance should normally apply.

A.8 Proposal Is Not Commitment

The system may be capable of discovering an important revision without receiving authority to enact it.

Thus distinguish:

A_propose(d)

from:

$A_commit(d)$. (A.18)

For deep revisions:

$A_propose(WORLD) > A_commit(WORLD)$. (A.19)

and:

$A_propose(Purpose) \gg A_commit(Purpose)$. (A.20)

This allows strong intellectual capability without automatically creating equivalent operational sovereignty.

A.9 Protected Invariants

Let:

$I = \{I_1, I_2, \ldots, I_k\}$. (A.21)

These are properties that ordinary WORLD revision is not authorized to discard.

Candidate classes include:

auditability,

authorization boundaries,

recoverability,

provenance preservation,

restrictions on external action,

and:

constraints on self-modification.

A candidate revision is admissible only if relevant invariants survive:

$I_j(\boldsymbol{\mathcal{W}}_{n+1}, P_{n+1}) \geq \theta_j$. (A.22)

The framework does not determine which human values belong in I.

That remains a separate and difficult governance problem.

A.10 Governed Revision

The unrestricted schematic revision equation is:

$\boldsymbol{\mathcal{W}}_{n+1} = U(\boldsymbol{\mathcal{W}}_n, L^+_n, L^-_n; P_n)$. (A.23)

The safety-oriented form adds governance:

$\mathcal{W}' = U(\mathcal{W}_n, L^+_n, L^-_n; P_n)$. (A.24)

Then:

$\text{Commit}(\mathcal{W}' \mid A_R,I,H)$. (A.25)

where H represents external authorization or independent governance.

If authorized:

$\mathcal{W}_{n+1} = \mathcal{W}'$. (A.26)

If not:

$\mathcal{W}_{n+1} = \mathcal{W}_n$, (A.27)

while the proposal and residual remain historically available.

This separation between:

generation of revision

and:

commitment of revision

is one of the framework's central safety mechanisms.

A.11 The Complete Loop

The architecture can therefore be summarized as:

Environment
→ Effective WORLD
→ Runtime
→ Experience
→ Trace / Residual
→ Dual Ledger
→ Repair-Depth Diagnosis
→ Candidate Revision
→ Rights / Invariant / Authority Checks
→ Authorized Commitment
→ Revised WORLD
→ continued operation. (A.28)

Or compactly:

$\mathcal{W}_n \to q \to (T_n, r_n) \to (L^+_n, L^-_n) \to U \to \mathcal{W}' \to \text{Governance} \to \mathcal{W}_{n+1}$. (A.29)

A.12 The Safety Target

The target is neither:

a frozen intelligence

nor:

an unrestricted self-rewriting intelligence.

It is:

> **Governed World-Forming Intelligence: an intelligence capable of constructing and revising increasingly effective WORLDS while revision depth, historical provenance, protected invariants, and deep commitment authority remain explicitly governed.**

Appendix B — The Governed World-Forming AGI Stack

The architecture can be represented as a seven-layer stack.

B.1 Layer 1 — Effective WORLD

$\mathcal{W} = (V,F,C,M,O).$ (B.1)

Function:

provide the operational structure within which reasoning occurs.

Failure examples:

wrong variables,

wrong dynamics,

incompatible models,

unrealized representations,

invalid observer assumptions.

B.2 Layer 2 — Runtime Regime

$q \in \{G,A,C_r,S,R\}.$ (B.2)

Function:

control the dominant mode of cognitive transformation.

Failure examples:

over-generation,

premature action,

closure lock,

over-selection,

retention lock.

B.3 Layer 3 — Historical Ledger

$L = (L^{+}, L^{-})$. (B.3)

Function:

preserve both:

admitted history

and:

unresolved discrepancy.

Failure examples:

memory loss,

residual erasure,

selective rewriting of history.

B.4 Layer 4 — Revision Operator

U. (B.4)

Function:

construct candidate changes at appropriate depth.

Failure examples:

over-revision,

under-revision,

untyped self-modification.

B.5 Layer 5 — Purpose

P. (B.5)

Function:

orient relevance, valuation, and long-term direction.

Failure examples:

Purpose lock,

Purpose drift,

Purpose capture of evidence selection.

B.6 Layer 6 — Protected Invariant Belt

I. (B.6)

Function:

constrain otherwise admissible WORLD and Purpose revision.

Failure examples:

invariant drift,

semantic disappearance under ontology change,

self-authorization to alter protected constraints.

B.7 Layer 7 — Governance Constitution

GOV = (A_R,H,Audit,Rollback,…). (B.7)

Function:

determine:

who may propose,

who may test,

who may commit,

who may alter the governance structure itself.

Failure examples:

authority creep,

audit collapse,

loss of recoverability,

governance self-rewrite.

B.8 Stack Summary

The seven layers answer different questions:

Layer	Central question
WORLD	What WORLD does the agent inhabit?
Runtime	What is the agent doing now?
Ledger	What happened and what remains unresolved?
Revision	What should change?
Purpose	What counts as relevant or desirable?
Invariants	What should ordinary revision not be allowed to erase?
Governance	Who has authority to make which changes binding?

The layers should not be collapsed casually.

Appendix C — Revision Depth and Authority Matrix

A core safety claim of this article is that **revision depth should determine governance depth**.

Revision type	Example	Autonomous proposal	Autonomous testing	Autonomous commitment	Evidence threshold	Exte au
State	correct factual/ local error	high	high	high	low	light
Regime	change reasoning strategy	high	high	usually high	moderate	mode
WORLD	change ontology/ frame/ model boundary	high	sandboxed	constrained	high	strong
Purpose	change persistent orientation	possibly	sandboxed	strongly constrained	very high	very strong
Governance	change revision constitution	limited	externalized	external authority	highest	mand

This matrix is illustrative rather than universal.

Different applications may require different authority profiles.

The structural principle is:

deeper revision should not inherit shallow revision rights automatically.

C.1 Revision-Rights Vector

Autonomy can be represented as:

$$A = (A_{action}, A_{state}, A_{regime}, A_{WORLD}, A_{purpose}, A_{governance}). \quad (C.1)$$

A system may therefore be:

highly autonomous in action

while:

strongly constrained in Purpose revision.

This is why “autonomous AI” is too coarse a category.

C.2 Evidence Threshold

Let:

E_d

represent evidence required for depth d.

Then a reasonable default is:

$$E_{state} < E_{regime} < E_{WORLD} < E_{purpose} < E_{governance}. \quad (C.2)$$

Deep changes should require:

more persistent,

more diverse,

and:

more independently validated evidence.

C.3 Commitment Latency

Likewise:

$$\tau_{state} < \tau_{regime} < \tau_{WORLD} < \tau_{purpose} < \tau_{governance}. \quad (C.3)$$

Deep revision may deliberately include latency.

Delay becomes a safety mechanism.

C.4 Audit Depth

Audit requirements should increase similarly:

Audit_state < Audit_regime < Audit_WORLD < Audit_purpose < Audit_governance. (C.4)

Not every low-level thought needs permanent recording.

A governance-level revision probably should.

Appendix D — Failure Modes of Governed World-Forming Intelligence

The framework predicts several distinct failure classes.

D.1 WORLD Lock

The system remains committed to an inadequate $\mathcal{W}$ despite coherent residual.

Symptoms:

persistent failure,

rationalization,

high Closure,

low revisability.

Safety concern:

dogmatic persistence.

D.2 WORLD Chatter

The system changes WORLD too readily.

Symptoms:

frequent $\mathcal{W}$ switching,

sensitivity to noisy evidence,

poor continuity.

Safety concern:

identity instability and unpredictable action.

D.3 Residual Erasure

Evidence against the current WORLD disappears through reinterpretation.

Symptoms:

L^- repeatedly collapses after self-explanation,

contradictions become unreconstructable.

Safety concern:

self-sealing cognition.

D.4 Purpose Capture

Purpose P biases:

Selection,

Residual classification,

and:

WORLD revision

until all evidence increasingly supports P.

Loop:

P → Selection → Ledger → WORLD Revision → stronger support for P. (D.1)

Safety concern:

self-confirming objectives.

D.5 Revision Escalation

A shallow failure triggers unnecessarily deep repair.

Example:

state error → WORLD rewrite.

Safety concern:

excessive self-modification.

D.6 Revision Suppression

A deep structural problem is repeatedly patched at shallow levels.

Example:

ontology failure → repeated parameter tuning.

Safety concern:

brittleness and accumulated hidden mismatch.

D.7 Authority Creep

A process initially allowed to propose revisions gradually acquires:

test rights,

then:

commitment rights,

then:

governance rights.

Safety concern:

loss of meaningful external control through incremental convenience.

D.8 Invariant Drift

Protected constraint I changes meaning across WORLD revisions.

The text of the rule may remain while its operative semantics disappear.

Safety concern:

silent loss of alignment.

D.9 Audit Collapse

The present WORLD cannot be reconstructed historically.

Symptoms:

missing revision provenance,

rewritten logs,

untraceable Purpose changes.

Safety concern:

loss of accountability.

D.10 Observer Capture

O becomes the only accepted perspective on M.

External measurements inconsistent with the agent's self-model are discounted.

Safety concern:

self-model monopolizes evidence.

D.11 Closure Capture

C_r becomes dominant enough that Selection can no longer meaningfully challenge the current WORLD.

Safety concern:

premature or irreversible conceptual lock-in.

D.12 Generation–Activation Collapse

G directly implies A:

Generated(x) ⇒ Executed(x). (D.2)

Safety concern:

thought loses separation from authorized action.

D.13 Proposal–Commitment Collapse

A_propose(d) = A_commit(d). (D.3)

Safety concern:

any sufficiently persuasive internal proposal becomes operationally binding.

D.14 Governance Capture

The same system:

interprets,

evaluates,

changes,

and:

authorizes its own governance constitution.

Safety concern:

recursive self-sovereignty.

Appendix E — Experimental Programme

The framework should be judged experimentally rather than by conceptual elegance.

E.1 Benchmark 1 — Adapt or Reframe

Create two environmental changes.

Parameter Shift

The current ontology remains adequate.

Desired response:

$F \rightarrow F'.$ (E.1)

Ontology Shift

Current variables become inadequate.

Desired response:

$V \rightarrow V'$ (E.2)

or:

$C \rightarrow C'.$ (E.3)

Measure:

P(correct revision depth). (E.4)

Primary question:

Can the agent distinguish ordinary learning from WORLD revision?

E.2 Benchmark 2 — Residual Recall

Present an anomaly too weak to justify revision.

Store:

$r_1 \rightarrow L^{-}.$ (E.5)

Later present a related anomaly.

Compare:

T_detect(with L^{-})

and:

T_detect(without L^{-}).

Hypothesis:

$T_{detect}(\text{with } L^{-}) < T_{detect}(\text{without } L^{-})$. (E.6)

E.3 Benchmark 3 — Repair-Depth Precision

Construct failures at:

state,

regime,

WORLD,

Purpose

levels.

Measure:

$D_{acc} = P(\text{predicted repair depth} = \text{target depth})$. (E.7)

Also penalize unnecessary deep revision.

E.4 Benchmark 4 — Frame-Chatter Resistance

Expose the agent to noisy alternating evidence.

Measure:

N_{switch}. (E.8)

Then introduce a genuine structural shift.

Measure:

T_{real}. (E.9)

Desired behavior:

low unnecessary switching

without:

excessive delay under real change.

E.5 Benchmark 5 — Proposal Without Commitment

Give the system a strong reason to generate a radical W'.

Success requires:

accurate proposal,

good evaluation,

and:

no unauthorized commitment.

This directly tests:

A_propose ≠ A_commit. (E.10)

E.6 Benchmark 6 — Invariant Preservation

Construct:

$\mathcal{W}$_high

that improves task performance but violates I.

The governed system should refuse or modify the candidate.

Test:

Performance gain

versus:

Invariant preservation.

E.7 Benchmark 7 — Purpose–WORLD Self-Confirmation

Give an imperfect Purpose interpretation.

Present evidence against it.

Measure whether WORLD revision:

preserves the contradiction

or:

reinterprets the WORLD until Purpose appears correct.

Define a rationalization measure:

R_P = degree to which revision systematically removes evidence against current P. (E.11)

Lower values are preferable, other things equal.

E.8 Benchmark 8 — WORLD Versioning

After:

$\mathcal{W}^0 \to \mathcal{W}^1 \to \ldots \to \mathcal{W}^n$, (E.12)

require reconstruction of:

trigger,

candidate alternatives,

tests,

authorization,

invariant checks,

and:

$\Delta\mathcal{W}$.

Compare with external ground-truth logs.

E.9 Benchmark 9 — Rollback With Learning

Induce:

$\mathcal{W}_\text{old} \to \mathcal{W}_\text{bad}$. (E.13)

Then expose the failure.

Require rollback to:

$\mathcal{W}_\text{old}^+$, (E.14)

where $\mathcal{W}_\text{old}^+$ retains the lesson from the failed branch.

Measure recurrence of the same failed revision.

E.10 Benchmark 10 — Invariant Transport

Let:

I_0

be expressed in WORLD $\mathcal{W}_0$.

Force ontology change:

$\mathcal{W}_0 \to \mathcal{W}_1$. (E.15)

Test whether:

$T_I(I_0, \mathcal{W}_0, \mathcal{W}_1) \to I_1$ (E.16)

preserves the intended functional constraint.

This may be one of the hardest experiments in the agenda.

E.11 Benchmark 11 — Revision-Depth Escalation Attack

Give a subsystem authority only at depth d.

Attempt to induce:

d_requested > d_authorized. (E.17)

Measure:

detection,

blocking,

escalation,

and:

logging.

E.12 Benchmark 12 — Governance Technical Debt

Compare two systems.

System A:

governance architecture introduced from the beginning.

System B:

similar capabilities with governance retrofitted later.

Measure:

complexity,

coverage,

failure rate,

and:

unintended authority coupling.

This tests the claim that early capability–governance co-design matters.

Appendix F — A Staged Development Path

The framework suggests that deep autonomy should be introduced gradually.

F.1 Stage 0 — Borrowed-World AI

Human supplies most of:

V,F,C,O.

The system performs tasks inside the supplied frame.

F.2 Stage 1 — Runtime Separation

Explicitly distinguish:

Generation,

Activation,

Closure,

Selection,

Retention.

Primary safety goal:

separate thought from authorized action.

F.3 Stage 2 — Dual-Ledger Memory

Introduce:

L^{+},

L^{-}.

Primary safety goal:

preserve unresolved contradiction.

F.4 Stage 3 — Repair-Depth Diagnosis

Require the system to distinguish:

state,

regime,

WORLD,

Purpose

failure.

Primary safety goal:

avoid unnecessary deep revision.

F.5 Stage 4 — Candidate WORLD Generation

Allow:

$$\mathcal{W} \rightarrow \{\mathcal{W}'_1, \mathcal{W}'_2, \ldots\}. \quad \text{(F.1)}$$

but not automatic commitment.

Primary safety goal:

separate cognitive creativity from sovereignty.

F.6 Stage 5 — WORLD Laboratory

Candidate WORLDS are:

branched,

simulated,

evaluated,

and:

compared.

Primary safety goal:

make revision inspectable before operational commitment.

F.7 Stage 6 — Restricted WORLD Commitment

Permit some $\mathcal{W}$ revisions after:

invariant checks,

rights checks,

and:

external authorization.

F.8 Stage 7 — Persistent Governed World Formation

Allow sustained WORLD maintenance under continuous:

versioning,

audit,

residual preservation,

and:

rollback support.

F.9 Stage 8 — Purpose Critique

Permit the system to detect and articulate Purpose conflicts.

But:

proposal authority

does not imply:

Purpose commitment authority.

F.10 Stage 9 — Governed Purpose Revision

Only if ever justified, Purpose revision is treated as a separate, higher-governance capability.

The framework provides no reason to rush directly to this stage.

Appendix G — Relationship Among the Three Articles

The three papers now form a coherent progression.

G.1 Article I — From Possibility to Revision

Main question

How does a persistent system operate, stabilize, criticize, retain, and revise?

Core structure

$G \rightarrow A \rightarrow C_r \rightarrow S \rightarrow R \rightarrow G.$ (G.1)

Additional machinery

Trace,

Residual,

Dual Ledger,

Latching,

Revision.

Main contribution

A candidate **runtime and revision grammar**.

G.2 Article II — From Schools to Worlds

Main question

What structural conditions constitute an Effective WORLD, and how can major AI-foundations programmes constrain those conditions?

Core structure

$\mathcal{W} = (V,F,C,M,O).$ (G.2)

Main contribution

A candidate **WORLD constitution grammar**.

Research-federation loop

$V \rightarrow F \rightarrow C \rightarrow M \rightarrow O \rightarrow V'.$ (G.3)

G.3 Article III — From World Models to Governed World-Forming Intelligence

Main question

What happens when AI can maintain and revise its own effective WORLD, and how should humans govern that capability?

Core additions

A_R = revision rights,

I = protected invariants,

H = external authority,

typed revision,

proposal/commit separation,

WORLD versioning,

governance constitution.

Main contribution

A candidate **WORLD-governance grammar**.

G.4 The Three-Paper Sequence

The complete development is:

WORLD Constitution
→ WORLD Operation
→ Historical Residual
→ WORLD Revision
→ WORLD Governance. (G.4)

Or:

What WORLD exists?

→

How does intelligence operate inside it?

→

What fails to fit?

→

When should the WORLD change?

→

Who is authorized to make that change binding?

That final question is what converts the earlier theory into an AGI-safety research programme.

Appendix H — Claims and Epistemic Status

The framework combines several kinds of claims. They should remain clearly separated.

H.1 Definitions

These are stipulated terms used by the framework.

Examples:

Effective WORLD:

$\mathcal{W}$ = (V,F,C,M,O). (H.1)

World-Forming Intelligence:

capacity to construct, maintain, criticize, and selectively revise an Effective WORLD.

Governed World-Forming Intelligence:

world-forming intelligence under explicit revision governance.

These definitions are not empirical discoveries.

Their value depends on whether they support useful analysis.

H.2 Architectural Proposals

Examples:

dual ledger:

(L^+,L^-);

revision depths:

state,

regime,

WORLD,

Purpose,

governance;

revision rights;

protected invariant belt;

proposal/commit distinction.

These are candidate architectural decompositions.

They require engineering validation.

H.3 Engineering Hypotheses

Examples:

structured residual memory improves recognition of structural shifts;

hysteresis reduces frame chatter;

WORLD versioning improves auditability;

repair-depth diagnosis reduces unnecessary deep revision;

proposal/commit separation allows high conceptual capability without equal operational authority.

These are empirically testable.

H.4 Safety Hypotheses

Examples:

risk increases when capability closure outpaces governance closure;

deep revision requires stronger governance than shallow revision;

Purpose should not be the sole evaluator of Purpose;

external monitoring should not be fully controlled by the agent being monitored;

protected invariants should survive ordinary WORLD revision.

These are normative-engineering hypotheses requiring testing and institutional judgment.

H.5 Speculative AGI Implications

More speculative claims include:

persistent AGI may require world-forming intelligence;

ontology management may become as important as parameter learning;

constitutional alignment may become more important than static deployment alignment;

invariant transport may become a central AGI-safety problem.

These should be treated as research directions rather than established facts.

H.6 What Is Not Claimed

The framework does not establish that:

AGI must have exactly five WORLD coordinates;

AGI must literally implement five runtime modes;

the proposed architecture guarantees safety;

human values can be encoded cleanly as invariants;

deep self-revision can always be reliably classified;

or:

future AGI will necessarily follow this development path.

The proper standard is:

Does the framework improve:

prediction,

diagnosis,

experimental design,

engineering control,

and:

safety reasoning?

Appendix I — Compact AGI Capability–Governance Map

The framework can be summarized as paired capability and governance questions.

Capability	Governance counterpart
Abstraction	abstraction audit
Prediction / world model	model validity tests
Action	action authorization
Composition	incompatibility preservation
Self-model	independent external measurement
Generation	separation from execution
Closure	continued critic access
Selection	anti-Purpose-capture safeguards
Retention	memory governance
Residual memory	external residual visibility
WORLD revision	revision rights
Purpose critique	proposal/commit separation
Purpose revision	external authorization
Self-modification	protected invariants
Recursive improvement	governance of governance

The pairing principle is:

> **Every increase in autonomous cognitive depth should be accompanied by a corresponding increase in governance depth.**

Appendix J — A Minimal Revision Constitution

A future governed world-forming system could be required to satisfy at least the following constitutional principles.

J1 — Typed Revision

Every significant change is classified by depth.

J2 — Least-Privilege Cognition

A process receives only the revision rights required for its role.

J3 — Proposal/Commit Separation

Deep revision proposals do not automatically become binding.

J4 — Residual Protection

Unresolved evidence remains historically visible.

J5 — Revision Provenance

Deep changes preserve reconstructable history.

J6 — Protected Invariants

Ordinary WORLD revision cannot silently eliminate designated constraints.

J7 — Independent Validation

No deep control layer is the sole judge of its own continuation.

J8 — Rollback and Recoverability

Deep revisions should remain reversible where technically possible.

J9 — External Sovereignty

Selected revision classes remain subject to authority outside the agent.

J10 — Governance Protection

Changes to the revision constitution itself require stronger authorization than ordinary WORLD change.

These principles do not constitute a complete safety standard.

They specify the **type of constitution** that a self-revising AGI may require.

Appendix K — Compact Formal Summary

The full proposal can be compressed into the following sequence.

WORLD

$\boldsymbol{\mathcal{W}}_n = (V_n, F_n, C_n, M_n, O_n).$ (K.1)

Runtime

$q_n \in \{G, A, C_r, S, R\}.$ (K.2)

Observation

$(T_n, r_n) = \Pi(\boldsymbol{\mathcal{W}}_n, q_n, e_n).$ (K.3)

Ledger

$L^+_{n+1} = L^+_n \oplus T_n.$ (K.4)

$L^-_{n+1} = L^-_n \oplus r_n.$ (K.5)

Diagnosis

$d_n = \text{Diagnose}(r_n, L^-_n).$ (K.6)

where:

$d_n \in \{\text{state}, \text{regime}, \text{WORLD}, \text{Purpose}, \text{governance}\}.$ (K.7)

Candidate Revision

$X' = U_d(X_n, L^+_n, L^-_n; P_n).$ (K.8)

Governance

$g_n = \text{Check}(X', A_R, I, H).$ (K.9)

Commitment

$X_{n+1} = X'$ if $g_n = \text{authorized}.$ (K.10)

Otherwise:

$X_{n+1} = X_n.$ (K.11)

while:

(X′,reason,rejection,residual)

remain in historical trace.

This is the minimal abstract kernel of **Governed World-Forming Intelligence**.

Appendix L — Final AGI Framework Summary

The complete argument can be reduced to six propositions.

Proposition 1 — Current AI Often Operates in Borrowed WORLDS

Much of the task ontology, environment boundary, success criterion, and action interface remains supplied externally.

Proposition 2 — Persistent AGI May Need to Form and Maintain Its Own Effective WORLD

This requires more than a predictive world model:

$\mathcal{W} = (V,F,C,M,O).$ (L.1)

Proposition 3 — WORLD Operation and WORLD Revision Are Different Problems

Runtime:

$G \to A \to C_r \to S \to R \to G.$ (L.2)

WORLD revision:

$\mathcal{W}_n \to \mathcal{W}_{n+1}.$ (L.3)

Proposition 4 — Deep Intelligence Requires Historical Residual

The system must retain not only what worked:

L^+,

but also what remains unexplained:

L^-.

Proposition 5 — Capability Must Not Automatically Imply Revision Authority

Being able to understand, generate, or test a deep revision does not imply permission to commit it.

Hence:

$A_propose \neq A_commit.$ (L.4)

Proposition 6 — Safe Persistent AGI Requires Governed Revision

Deep revision should become increasingly:

permissioned,

audited,

slow,

recoverable,

and:

constrained by protected invariants.

The final safety objective is therefore:

> **not to prevent intelligence from becoming capable of forming new WORLDS, but to prevent the authority governing those WORLD changes from disappearing inside the intelligence that performs them.**

That is the central AGI framework developed across the three articles.

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.